DYSANOS

Generative Dynamic Smooth Arbitrage-free Non-parametric Option Surfaces

Hans Buehler* Blanka Horvath† Anastasis Kratsios‡ Magnus Wiese§

October 3, 2026
Version 2.0

Abstract

This article presents with DYSANOS the first generative market model for smooth SANOS option surfaces for all strikes and expiries which are free of static arbitrage. Our model is designed to generate entire paths of daily spot and option prices for years in the future.

We present a robust and useful if somewhat simplistic baseline in the form of an AR(1) model. We discuss model setup, data pipeline, and training and investigate market reconstruction, stability, and tail behavior. We illustrate model performance on 891 Option Metrics IvyDB S&P Index surfaces from 2022 to 2025, drawn from an underlying archive beginning in 2020. With explicit 1D–4D expiry nodes, the two-η affine-200 model places 80.9% of 3.57 million quotes inside bid–ask. Finally, a no-replay scalar log-variance diffusion recovers much more of the observed tail mass and tail persistence than Gaussian innovations, although it still underproduces two- and three-week extremes. Block-randomization tests of decoded 2025 surface paths do not reject Gaussian, heavy-tail, diffusion, or historical-replay candidates at the 5% level. For a reduced, 45-instrument tradable universe we further construct, via Deep Hedging, a near-martingale measure whose empirically re-solved objective value is statistically indistinguishable from zero, a first, deliberately scoped step towards a risk-neutral DYSANOS.

Contents

1 Introduction **2**

2 DYSANOS **5**
2.1 Arbitrage-Free Call Price Functions 6
2.2 SANOS - Smooth Arbitrage-free Non-parametric Option Surfaces 7

3 ML-SANOS **10**
3.1 Learning ML-SANOS from real data 12
3.1.1 Market fit 13

4 Statistical option-surface state-space model **14**
4.1 Weekly AR(1) 14
4.2 Innovation laws and tail dynamics 16
4.2.1 No-replay stochastic-volatility diffusion 16
4.3 Chronological surface-path comparison 17
4.4 Stability diagnostics 19
4.5 Conditional front-expiry spot block 19

*Mathematical Institute and Oxford-Man Institute, University of Oxford. hans.buehler@maths.ox.ac.uk
†Mathematical Institute and Oxford-Man Institute, University of Oxford. blanka.horvath@maths.ox.ac.uk
‡McMaster University and Vector Institute.
§Mathematical Institute, University of Oxford. magnus.wiese@maths.ox.ac.uk

4.6 Decoded implied-volatility geometry . . . 20
4.6.1 Invariant-distribution visualization . . . 25
4.7 Simulation algorithm . . . 27
4.8 Fixed-sample to-expiry entropy projection . . . 27
4.9 Scope and limitations . . . 28

5 Risk-Neutral DYSANOS **29**
5.1 Universe and pre-conditioning . . . 29
5.2 A near-martingale measure via Deep Hedging . . . 29
5.3 Empirical confirmation . . . 30
5.4 Scope and limitations . . . 30

6 Conclusion **30**

7 Thanks **31**

A Appendix: Data Pipeline **31**
A.1 Pipeline Overview . . . 31
A.2 Raw Daily Market Data . . . 32
A.3 Enrichment and Expiry Filtering . . . 32
A.4 Arbitrage-Free Preprocessing . . . 33
A.5 Option Filtering . . . 33
A.6 Spot Normalization . . . 33
A.7 Arbitrage-free preprocessing and model geometry . . . 34
A.8 Fitting ML-SANOS . . . 34

1 Introduction

The implied volatility surface of an option market changes dynamically over time, giving rise to Vega risk when risk managing option portfolios. Modelling the dynamics of a full continuous surface of options in a way that is simultaneously *statistically realistic* and *statically arbitrage-free* at every simulated date has remained an open problem since the foundational work of [CFD02] on dynamics of the implied volatility surface.

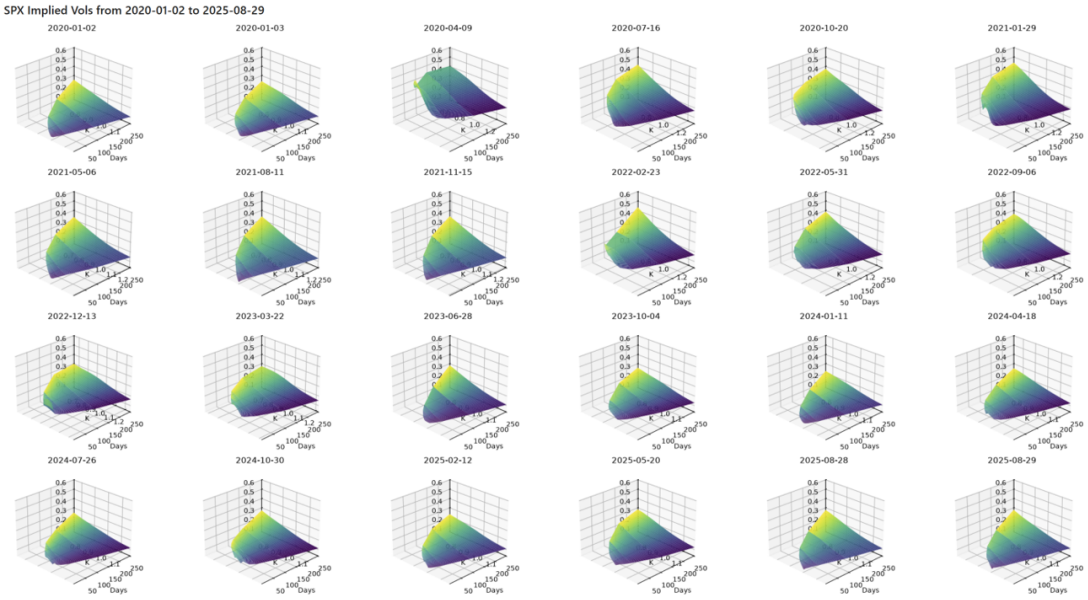


Figure 1: Historical implied volatilities for mid-prices of S&P 500 Index options from 2020-01-02 to 2025-08-29 using Option Metrics Ivy DB data via WRDS https://wrds-www.wharton.upenn.edu/ (the latest data available at the time of writing).

The most advanced approach until recently were the simulators [WBWB19, WWP+21, CRW23] for realistic dynamics of a *discrete floating grid* of options which are strictly arbitrage-free at each simulation date. These simulators are using, in essence, the discrete local volatility representation pioneered in [BR15] to ensure absence of static arbitrage in every simulated option surface.

However, the restriction of the above models to a discrete set of floating option strikes and time-to-expiries means that the simulators do not have access to the daily returns $dC_t(\tau, k)$ of an actual tradable instruments with cash strikes k and calendar expiries τ (because the contract terms change from one simulated day to another). It therefore does not directly provide statistically consistent time series of fixed-contract returns, which are required to validate simulated dynamics against observed market returns. Of course, since the discrete option prices are free of arbitrage, one option would be to interpolate option prices linearly, which means all options are free of static arbitrage. However, as shown in [Bue06] and illustrated in [BHK+26] linear interpolation yields the most expensive possible arbitrage-free price for any option off the grid, rendering the approach unsuitable in practice. Being linear also means that the implied probability density is piecewise constant and zero between strikes. In other words the spot price can only take values at the model strikes for each expiry to avoid dynamic arbitrage.

This article presents a model framework for

- **(i) realistic generative dynamics** of
- **(ii) smooth option surfaces for all strikes and expiries** which are
- **(iii) free of static arbitrage**.

We also comment on finding a risk-neutral measure for a subset of traded options. This article is a refinement of material presented at The XIII Bachelier World Congress in Bologna.[1]

[1] https://quantitative-research.de/dl/DYSANOS.pdf

Historic Context and Literature Review

The classical approach and its limitations. Quantitative finance has traditionally approached this problem by specifying analytically tractable risk-neutral dynamics for the underlying asset — Heston, rough volatility, SABR — and then *calibrating* model parameters to fit each day's option surface independently. They are then overlaid with a local volatility surface using the seminal [GHL12]. The latter guarantees that all options are initially priced correctly.

In this paradigm, realistic surface dynamics are an afterthought: calibration is repeated from scratch on each date. This approach is both impractical for large-scale simulation and structurally inconsistent with the goal of generating realistic implied volatility or option price paths.

Simulating Implied Volatility. A natural data-driven alternative is to treat the implied volatility surface itself as the simulation object, much as implied in [CFD02]. The fundamental difficulty is that there is no known parametrization of the implied volatility surface which is strictly free of static arbitrage. There exists numerous attempts to rectify this with classic (e.g. [Fen09]) and machine learned methods (e.g. [NJZB23]) but to our knowledge none of them can strictly guarantee absence of static arbitrage.

If a simulated surface admits a butterfly spread or calendar spread arbitrage, then a learning agent can exploit an artificial profit which is unrelated to the problem for which the generated market was intended. Aside from those classic constraints call prices also need to adhere to boundary conditions which are often not discussed in the literature (we repeat them below).

Most published works also simplify the problem by only considering few strikes and expiries.

Constrained Parameter Spaces and Code Books. The core challenge of any representation of an option prices surface is that no-static-arbitrage conditions are essentially linear conditions on call prices, which are difficult to enforce directly in a generative machine learning model. This challenge is shared by most representations including most parametrizations of implied volatility.

The idea of a "code book" as introduced in [CN12] is to find a representation of the option price surface which is numerically tractable for simulation and for which required parameter constraints are straight forward to enforce. Indeed, the seminal [CN12] achieves that goal. In addition, the authors are able to also construct a risk-neutral measure consistent with the code book representation. They achieve this, essentially, by writing the spot price as a product of independent increments. That construction unfortunately also restricts the dynamics their model can represent. It tends to fit poorly to statistical data due to the strong cross-expiry correlation of implied forward volatilities.

Local Volatility as a Code Book. An intuitive alternative is to use Dupire's local volatility as "code book" instead *(not as dynamical model)*. Given a non-negative local volatility function $\sigma_t(T,K)$ for $(T,K) \in \mathbb{R}^2_{\geq 0}$ defined on some day t we can define a smooth, strictly arbitrage-free option surface $C_t(T,K) := C^{\mathrm{LV}}(\sigma_t;T,K)$ by solving the Dupire forward-PDE $\partial_T C^{\mathrm{LV}}(\sigma_t;T,K) = \frac{1}{2}K^2\sigma_t^2(T,K)\,\partial^2_{KK}C^{\mathrm{LV}}(\sigma_t;T,K)$. That means any statistical generative model of Dupire's local volatility gives rise to a dynamic model of option surfaces which are by construction free of static arbitrage.

However, in order to price any simulated option off a simulated local volatility surface we would need to solve a forward-PDE on a suitably fine grid which is slow, particularly when used within batch-based gradient descent training.

The idea to address this conundrum is to use a grid of options and define a discrete version of local volatility on that grid, which is what we called *discrete local volatility (DLV)* in [BR15]. It is essentially the discretized local volatility which is consistent with using an implicit forward-PDE operator on a sparse inhomogeneous grid of options, as explained in [AH11]. However, its native

option prices for strikes not on its base grid are the result of linear interpolation and therefore, as mentioned above, "too expensive". We therefore use our smooth SANOS extension [BHK+26].

The use of SANOS also allows to fit our model very efficiently to observed bid/ask spreads over the entire observed option surface.

Key contributions

SANOS Decoder:

We show how to implement and learn a coordinate transformation of our SANOS [BHK+26] model such that the "ML-SANOS" state parameters h_t live in an unconstrained space $\mathbb{R}^d$. The call prices

$$h_t \longmapsto C(h_t; T, K) \ .$$

remain smooth and statically arbitrage-free.

We show how ML-SANOS can be trained directly to suitably pre-processed option market data, yielding a trained h_t for each day. The pre-processing step means that DYSANOS, via SANOS, is fitted to real market bid/ask prices.

Generative Dynamics

We fit a stable weekly AR(1) directly to the learned surface state. Its transition is parameterized as the matrix exponential of a dissipative generator, and its full innovation covariance is learned by a nonsingular Cholesky factor. The resulting stationary law permits exact Gaussian transitions at daily, weekly, or longer horizons; as non-Gaussian alternatives, a no-replay log-variance diffusion is fitted and historical one-week innovations can be resampled.

We do not claim that this approach provides the most realistic dynamics; rather, we see this implementation as baseline to beat with more modern time series models.

Risk-Neutral Dynamics

Every decoded SANOS surface is free of static arbitrage by construction. We show that we can "remove the drift" in the sense of [BMPW22a] for a subset of the (infinite) tradable instruments by finding a near-martingale measure $\mathbb{Q}$ equivalent to the learned $\mathbb{P}$ such that a Deep Hedging [BGTW19] agent cannot make any money trading under $\mathbb{Q}$.

2 DYSANOS

We consider European call options on a single underlying for historic days $t \in \{t_1, \dots, t_m\}$. We assume forwards, discount factors, and survival probabilities are deterministic. As described in remark 2.1. we can therefore convert them at each point t to driftless equivalents where spot, forwards, discount factors and survival probabilities are all one, for every t.

At time t, we observe $L_t \in \mathbb{N}$ European call option prices with time-to-expiries $T_t^\ell \geq 0$ and relative strikes $K_t^\ell \geq 0$ for $\ell = 1, \dots, L_t$ in the market. Their symbolic payoff is therefore

$$\left(\frac{S_{t+T_t^\ell}}{S_t} - K_t^\ell \right)^+ \ .$$

We denote by B_t^ℓ and A_t^ℓ their bid and ask prices; mid-price and half-spread are defined (in vector notation) as $C_t := \frac{1}{2}(A_t + B_t)$ and $\gamma_t := \frac{1}{2}(A_t - B_t)$.

We generally assume that $B_t \succ 0$ and $\gamma_t \succ 0$.[2] Contrary to most derivative literature, we do not assume that bid/ask prices are in some way free of arbitrage; instead we will show how to handle arbitrage valuations gracefully.

[2] We define $x \succ y \Leftrightarrow \forall i : x^i > y^i$.

We will use the Black & Scholes formula in terms of *total volatilities* defined for a spot S, strike K and total volatility W as

$$\text{BSCall}(S, K, W) := S\,\mathcal{N}(d_+) - K\,\mathcal{N}(d_-) \quad \text{for} \quad d_\pm := \frac{1}{W}\log K/S \pm \frac{1}{2}W^2 \tag{1}$$

where $\mathcal{N}$ denotes the normal distribution function.

Remark 2.1 (Conversion to driftless data)**.** *Assume that at each time t spot is S_t, forwards are $F_t(T)$, discount factors are $B_t(T)$, and survival probabilities are $Q_t(T)$. It was shown in [Bue08] that if cash dividends are present then, conditional on no default, spot at T splits as $S_T = (F_t(T) - D_t(T))X_T + D_t(T)$ for a non-negative martingale X with $X_t = 1$, where $D_t(T)$ represents the discounted floor value of all cash dividends paid between t and T.*

We note that even in the presence of default risk put-call parity holds. Hence for any put $\tilde{P}_t(T, \tilde{K})$ with time-to-expiry T and cash strike $\tilde{K}$ let $\tilde{C}_t(T, \tilde{K}) = \tilde{P}_t(T, \tilde{K}) + B_t(T)\,Q_t(T)\,F_t(T) - B_t(T)\,\tilde{K}$.

Assume then that $\tilde{C}_t(T, \tilde{K})$ denotes the market call price for time-to-expiry T and cash strike $\tilde{K}$. Define the "pure" strike normalized to a unit forward and discount factor as

$$K := \frac{\tilde{K} - D_t(T)}{F_t(T) - D_t(T)}\ .$$

Accordingly, the "pure" call price is given as

$$C_t(T, K) := \frac{\tilde{C}_t\Big(T,\ K\,(F_t(T) - D_t(T)) + D_t(T)\Big)}{B_t(T)\,Q_t(T)\,(F_t(T) - D_t(T))}\ .$$

See [Bue08] for details and proof.

2.1 Arbitrage-Free Call Price Functions

A call price function is simply a function $C : \mathbb{R}^2_{\geq} \longrightarrow \mathbb{R}_{\geq 0}$ which computes call prices $C(T, K)$ for time-to-expiries $T \geq 0$ and relative strikes $K \geq 0$ (i.e. the call prices are normalized to a spot price of 1). The objective of this section and the next is to generate such functions C_t for each simulated date t such that each C_t is "free of static arbitrage" in the following sense:

Definition 2.2. *A call price surface $C : \mathbb{R}^2_{\geq 0} \to \mathbb{R}_{\geq 0}$ is* free of static arbitrage *if there is no trade which has non-negative returns with a non-zero probability of positive returns.*

This is equivalent to the existence of a martingale Z such that $C(T, K) = \mathbb{E}[(Z_T - K)^+]$.

Theorem 2.3. *A call price surface C is free of static arbitrage if and only if it does not admit negative butterfly or calendar spreads and satisfies the boundary conditions $C(T, 0) = 1$, $C(0, K) = (1 - K)^+$, $\lim_{K\uparrow\infty} C(T, K) = 0$ and $\partial_K C(T, 0) \in [-1, 0)$.*

The stricter condition $\partial_K C(T, 0) = -1$ applies if the stock price cannot become zero.

This has been proven in [Bue06]. Remark 2.4 shows that these conditions each are not superfluous: we give a practical guide on how to construct *actionable* arbitrage strategies if any of the above are violated.

Remark 2.4 (Actionable arbitrage if the conditions of theorem 2.3 are not met)**.** *All of the below are meant as practical heuristics.*

- ***Butterfly arbitrage:*** *this means that the price of a positive butterfly payoff is negative. The arbitrage is to get paid to enter into the butterfly which can only ever generate a positive return.*

- ***Calendar arbitrage:*** *if for the same strike K the call price at T_2 is below the call price at T_1, then:*

  1. *Sell the call at T_1 and buy the call at T_2. This provides a positive cash flow $\pi_0 > 0$.*
  2. *In T_1 if the call expiries out of the money, then we have made money: both π_0 and a potential positive payoff at T_2 in our favour.*

     If the call expires in the money, then we deliver one share and receive K. We are now short that share. In T_2, if the call expiries in the money, then we receive one share to cover our short in return for K. Our total cashflow is π_0. If the call expiries out of the money, then $S_T < K$. We can therefore buy back the share and cover our short for less than K. Our profits are $\pi_0 + (K - S_T)^+ > 0$.

- *If $C(0, K) \neq (1 - K)^+$ we have an immediate pathological arbitrage.*

- *If $C(T, 0) > 1$ then sell the call and cover with one share; if $C(T, 0) < 1$ then buy the call and enter into a short equity position.*

- *$\partial_K C(T, 0) \geq -1$: assume on the contrary there is a $K \downarrow 0$ such that $(1 - C(T, K))/(-K) < -1$ i.e. $C(T, K) = 1 - K - \epsilon$ for some $\epsilon > 0$. Buy the call and short sell the equity, resulting a cash position of $\pi_0 = -K + \epsilon$; the payoff in T is $(S_T - K)^+$ which allows us to cover our position and cash with a total reward of $\epsilon > 0$.*

- *$\partial_K C(T, 0) < 0$: assume on the contrary there is a $K \downarrow 0$ such that $(1 - C(T, K))/(-K) \geq 0$. We show existence of arbitrage in the case of equality i.e. existence of some $K > 0$ such that $C(T, K) = 1$. In that case we sell the call and buy the equity for 1. If the call expires in the money, we deliver the share and receive $K > 0$. If the call expiries out of the money, then sell the share for $S_T \geq 0$.*

- *$\lim_{K \uparrow \infty} C(T, K) = 0$ which means for all $c > 0$ there exists some K such that $C(T, K') < c$ for all $K' \geq K$ (because of convexity). This is the only arbitrage condition which is technical because a violation means, essentially, that the equity can grow without bounds which violates integrability. Assume therefore on the contrary that there is some $c > 0$ such that $C(T, K) \geq c$ for all K. Notice that, formally,*

$$S_T = -\int_0^\infty \frac{(S_T - K + dK)^+ - (S_T - K)^+}{dK} dK = S_T + \lim_{K \uparrow \infty} (S_T - K)^+ .$$

  Therefore sell (a discretization of) the strip

$$\int_0^\infty \frac{C(T, K) - C(T, K + dK)}{dK} dK = 1 + e$$

  which delivers S_T, and buy the equity. This will generate profits of $e > 0$.

2.2 SANOS - Smooth Arbitrage-free Non-parametric Option Surfaces

In [BHK+26] we introduced a non-parametric family of call price functions which are all statically arbitrage-free. For clarity, let us briefly fix t and drop it from the notation of market data and model parameters.

Each call price function is given in terms of model expiries $0 = \hat{T}^0 < \hat{T}^1 < \cdots < \hat{T}^M$ and, for each model expiry, $N^j > 0$ inner strikes.[3] Including the two boundary strikes, write $\hat{K}^j = (\hat{K}^{j,i})_{i=0,\ldots,N^j+1}$. It is recommended that the two outer strikes are far away from the

[3]Note that we changed notation slightly vs. [BHK+26].

inner strikes.[4] Model strikes and expiries do not need to match market strikes or expiries. See [BHK+26] for general comments on grid construction.

We say that $q = (q^1, \dots, q^M)$ with $q^j \in \mathbb{R}^{N^j+2}$ is a *martingale density* if it satisfies the linear constraints

$$\begin{cases} \text{Each } q^j \text{ is a density:} & q^j \geq 0,\ (q^j)'1 = 1 \\ \text{Each } q^j \text{ has unit expectation:} & (q^j)'\hat{K}^j = 1 \\ \text{The } q \text{ are in convex order:} & {q^j}'CX_j \geq {q^{j-1}}'CX_{j|j-1} \end{cases} \tag{2}$$

The matrices CX_j and $CX_{j|j-1}$ project the density on call prices. They are defined as

$$CX_j^{i,k} := (\hat{K}^{j,i} - \hat{K}^{j,k})^+ \quad i = 0, \dots, N^j + 1;\ k = 1, \dots, N^j$$

and

$$CX_{j|j-1}^{i,k} := (\hat{K}^{j-1,i} - \hat{K}^{j,k})^+ \quad i = 0, \dots, N^{j-1} + 1;\ k = 1, \dots, N^j\ .$$

Accordingly, ${q^j}'CX_j \in \mathbb{R}^{N_j}$ is the vector of call prices for strikes $\hat{K}^j \in \mathbb{R}^{N_j}$ computed using the density q^j while ${q^{j-1}}'CX_{j|j-1} \in \mathbb{R}^{N_j}$ is the vector of call prices for the same strikes $\hat{K}^j \in \mathbb{R}^{N_j}$ computed using the previous density q^{j-1}. The condition ${q^j}'CX_j \geq {q^{j-1}}'CX_{j|j-1}$ enforces that call prices are increasing into each strike $\hat{K}^j$.

Definition 2.5 (SANOS call price)**.** *Let q be a martingale density and $0 < W^1 < \dots < W^M$ be increasing smoothing total volatilities.*[5] *The SANOS call price for arbitrary $T \geq 0, K \geq 0$ is given by*

$$\begin{aligned} C(T,K) := \alpha^j(T) \sum_{i=0}^{N^j+1} \mathrm{BSCall}(\hat{K}^{j,i}, K, W(T)) q^{j,i} \\ + (1 - \alpha^j(T)) \sum_{i=0}^{N^{j-1}+1} \mathrm{BSCall}(\hat{K}^{j-1,i}, K, W(T)) q^{j-1,i}, \qquad T \in (\hat{T}^{j-1}, \hat{T}^j]. \end{aligned} \tag{3}$$

with $\alpha^j(T) = (T - \hat{T}^{j-1})/(\hat{T}^j - \hat{T}^{j-1})$ and total volatility $W(T) := \alpha^j(T)W^j + (1 - \alpha^j(T))W^{j-1}$.

For $T > \hat{T}^M$ we set $\alpha^{M+1}(T) = 0$ and $W(T) := W^M\sqrt{T/\hat{T}^M}$; however extrapolation beyond the last model expiry is generally not recommended.

For $T < \hat{T}^1$ we set $\alpha^1(T) = 0$ and $W(T) := W^1\sqrt{T/\hat{T}^1}$; however extrapolation before the first model expiry is generally not recommended.

Theorem 2.6. *The call price function C is statically arbitrage-free. For $W(T) > 0$ it is C^∞ in K. The function is as smooth as α in T.*

Remark 2.7. *Contrary to [BHK+26] we interpolated the Black & Scholes backbone in total volatility linearly, not variance. This is a mechanical choice to avoid taking square roots of variables during training which can lead to exploding gradients. If forward volatilities are bounded well away from zero, linear interpolation in variance is suitable, too.*

Remark 2.8. *The Black & Scholes formula in (3) can be replaced by the call price function for any martingale in T, for example the FFT Heston price formula [CM99]. For practical implementations, the function should be fast on a GPU.*

[4] In our experiments, the inner strikes $0 < \hat{K}^{j,1} < \dots < 1 < \dots < \hat{K}^{j,N^j}$ were chosen first and then extended by $\hat{K}^{j,0} := \hat{K}^{j,1}/2$ and $\hat{K}^{j,N^j+1} = 1.5\,\hat{K}^{j,N^j}$.

[5] The linear-programming preprocessing uses 50% of ATM total volatility. The trained ML-SANOS decoder instead learns the positive forward smoothing multipliers η^j.

Calibration as a linear program: We assume we obtain $W^1 < \cdots < W^M$ from the market e.g. as square roots of ATM variance per expiry. Given W the model call price $C^\ell := C(T^\ell, K^\ell)$ for a market expiry T^ℓ and a market strike K^ℓ can be seen in (3) to be linear in q. The market-fitting problem can therefore be formulated as the linear programme

$$\min_q \sum_\ell \frac{(C^\ell - \boldsymbol{A}^\ell)^+ + (\boldsymbol{B}^\ell - C^\ell)^+ + \epsilon|C^\ell - \boldsymbol{M}^\ell|}{(A^\ell - B^\ell)/2} \tag{4}$$

subject to the linear constraints (2) encoding non-negativity, normalisation, integration to 1, and the inter-expiry convex-order constraints that enforce absence of calendar-spread arbitrage. We note that the model strikes and time-to-expires do not need to correspond to market strikes or expiries.

As discusssed in [BHK+26] the primary application of this model is *interpolation*: if the model strikes are are superset of *all* strikes of relevant market instruments then the model can practically be fit to the vast majorty of options accross the entire surface from 1 day to expiries to several years. Stronger, if the model is linear ($W = 0$) the existence of the densities q is equivalent to absence of arbitrage with respect to the market bid/asks (usually this simply means our market data is not aligned correctly).

On S&P 500 Index data with 1000 options across 48 expiries (2-day to 657 business-day TTE), the LP solves in sub-seconds on a desktop PC, achieving 91.4% of options within bid–ask on a representative date (2025-05-06), with a median error of 21% of the half-spread on the remaining options. Figure 2 illustrates the quality of the fit. The example is from [BHK+26]; refer to that work for further details.

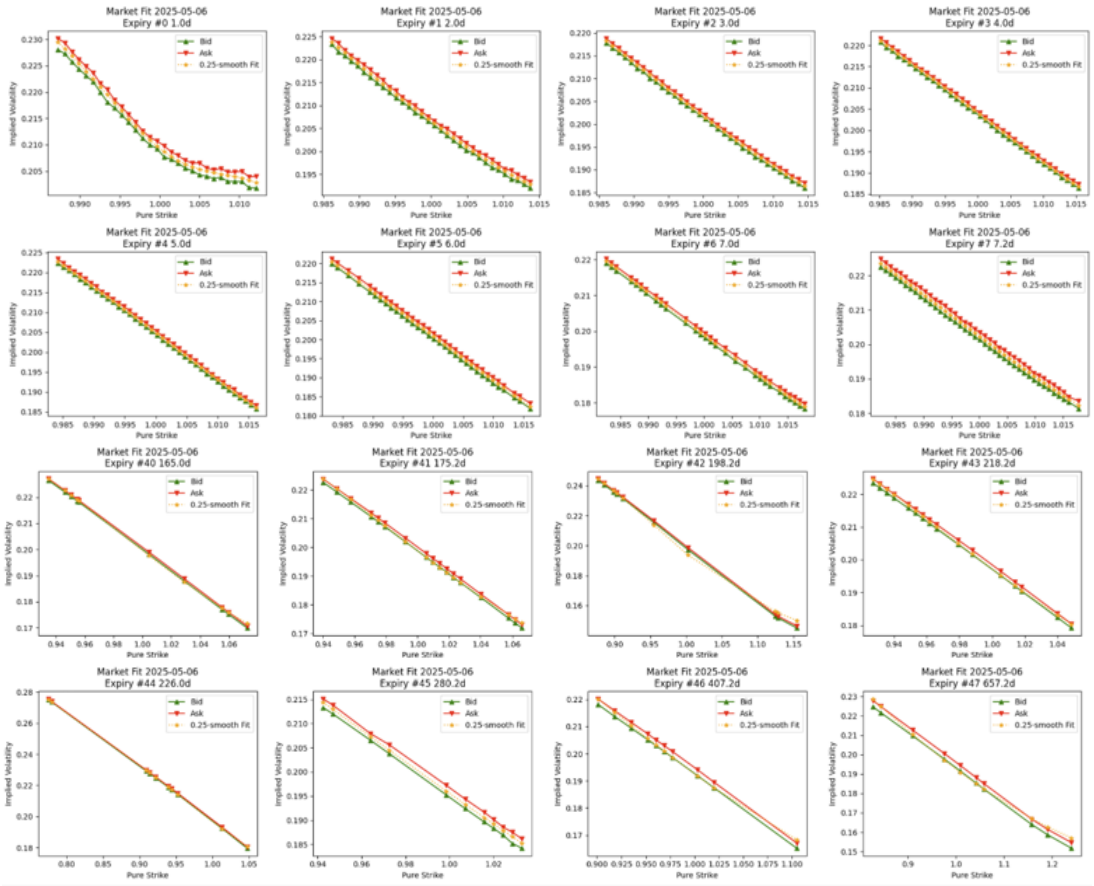


Figure 2: Example fit of SANOS.

3 ML-SANOS

We may view our SANOS call price operator as a map from the model parameters to a function space

$$(q, W;\, \hat{K}, \hat{T}) \overset{\text{SANOS}}{\longmapsto} C \tag{5}$$

for which we naturally want to learn q, W and, as we will see, $\hat{K}$ (as the right strike grid depends on the implied variance of the market). However, the density weights q^j in particular are subject to the linear constraints (2) for which no exact methods exist to integrate them into standard batch gradient descent methods. In this section we therefore reparameterize the map (5) in terms of learnable parameters.

To do so, we no longer require strict *interpolation* of the market prices. While we will want to be as close as possible to observed market prices (and will show that we are able to do so), we now focus on a model parametrization which lends itself to efficient learning using standard optimization techniques. That means that the number of strikes must somehow be kept manageable, rather than be dictated by the number of market instruments at each observed historic day t.

As a first step we therefore fix a common strike grid $\hat{K}$ with $0 < \hat{K}^0 < \cdots < 1 < \cdots < \hat{K}^{N+1}$ across expiries to simplify our model. The call price for strike $\hat{K}^i$ given the density q^j is given as $c^{j,i} = (q^j)'(\hat{K} - \hat{K}^i)^+$. The convex order condition in (2) therefore becomes simply $c^{j,i} \geq c^{j-1,i}$ for all i.

We now reparameterize our densities in terms of *discrete local volatilities (DLVs)* [BR15]: let

$$\omega^{i\pm} := \pm \frac{(\hat{K}^i)^2}{(\hat{K}^{i+1} - \hat{K}^{i-1})\,(\hat{K}^{i\pm1} - \hat{K}^i)} \geq 0$$

and with it the tridiagonal matrix Ω with

- Upper diagonal $(-\omega^{1-}, \ldots, -\omega^{N-}, 0)$,
- Lower diagonal $(0, -\omega^{1+}, \ldots, -\omega^{N^j+})$, and
- Main diagonal $(0, +\omega^{1+} + \omega^{1-}, \ldots, 1 + \omega^{N^j+} + \omega^{N^j-}, 0)$.

i.e.

$$\Omega = \begin{pmatrix} 0 & -\omega^{1-} & 0 & & & & \\ 0 & +\omega^{1+} + \omega^{1-} & -\omega^{2-} & & & & \\ 0 & -\omega^{1+} & +\omega^{2+} + \omega^{2-} & & & & \\ & & & \ddots & & & \\ & & & \ddots & & & \\ & & & & +\omega^{(N-1)+} + \omega^{(N-1)-} & -\omega^{N-} & 0 \\ & & & & -\omega^{(N-1)+} & +\omega^{N+} + \omega^{N-} & 0 \\ & & & & 0 & -\omega^{N+} & \end{pmatrix}. \tag{6}$$

Then for *any* squared (discrete) local volatilities $\Sigma^j = (0, \Sigma^{j,1}, \ldots, \Sigma^{j,N}, 0)$ the operator

$$1 + \Omega\ \mathrm{diag}(\Sigma^j)(\hat{T}_j - \hat{T}_{j-1}) \in \mathbb{R}^{(N+2)\times(N+2)} \tag{7}$$

is a tridiagonal operator with positive main diagonal and negative off diagonals. It is basically the discrete implicit scheme for the forward equation of the continuous time Dupire SDE. It is well known [BR15] that this operator is invertible, and the inverse Ξ^j can be efficiently computed on GPU using a compiled Thomas' algorithm. It is also a martingale transition operator in

the sense of [BR15] which means that the densities $q^1, \dots, q^M$ generated iteratively starting in $q^0 = (1)$ by

$$q^j = \Big(1 + \Omega \ \mathrm{diag}(\Sigma^j)(\hat{T}_j - \hat{T}_{j-1})\Big)^{-1} q^{j-1} \tag{8}$$

form a martingale density over $\hat{K}$.

With the above we have established a map from much simpler non-negative discrete local vols squared to densities, and then to SANOS call prices:

$$(\Sigma, W;\, \hat{K}, \hat{T}) \overset{\mathrm{DLV}}{\longmapsto} (q, W;\, \hat{K}, \hat{T}) \overset{\mathrm{SANOS}}{\longmapsto} C \ . \tag{9}$$

That allows us to pass to a practically unconstrained parameter set, which we call the ML-SANOS parametrization. We will use the sigmoid function $\mathrm{sigmoid}(x) = (1 + e^{-x})^{-1}$.

Definition 3.1 (ML-SANOS)**.** *Fix the normalized grid geometry for ML-SANOS with:*

- *Time-to-expiries (TTEs)* $0 = \hat{T}^0 < \cdots < \hat{T}^M$ *with* $\Delta\hat{T}^j = \hat{T}^j - \hat{T}^{j-1}$*;*
- *Normalized strikes* $z^1 < \cdots < 0 < \cdots < z^N$ *which we assume contain zero;*
- *Minimum and maximum forward volatilities* $\sigma_{\max} > \sigma_{\min} > 0$*;*
- *Minimum and maximum squared discrete local volatilities* $\Sigma_{\max} > \Sigma_{\min} > 0$*; and*
- *Minimum and maximum smoothing factors* $1 > \eta_{\max} > \eta_{\min} \geq 0$*.*

We translate the new real-valued ML-SANOS parameters $u = (u^\Sigma, u^\sigma, u^{\eta,\mathrm{f}}, u^{\eta,\mathrm{term}}) \in \mathbb{R}^{M\times N} \times \mathbb{R}^M \times \mathbb{R}^2$ *into SANOS parameters as follows:*

- *Total volatilities: define the* ***forward volatilities***

$$\sigma^j := \sigma_{\min} + \mathrm{sigmoid}(u^{\sigma,j})(\sigma_{\max} - \sigma_{\min}) \tag{10}$$

and two smoothing factors

$$\eta^{\mathrm{f}} := \eta_{\min} + (\eta_{\max} - \eta_{\min})\mathrm{sigmoid}(u^{\eta,\mathrm{f}}) \quad \textit{and} \quad \eta^{\mathrm{term}} := \eta_{\min} + (\eta_{\max} - \eta_{\min})\mathrm{sigmoid}(u^{\eta,\mathrm{term}}) \ . \tag{11}$$

With these quantities define the ***total volatilities***

$$\widetilde{W}^j := \sum_{\ell=1}^{j} \sigma^\ell \sqrt{\Delta\hat{T}^\ell} \qquad \textit{and} \qquad W^j := \eta^{\mathrm{f}} \sigma^1 \sqrt{\Delta\hat{T}^1} \ + \eta^{\mathrm{term}} \sum_{\ell=2}^{j} \sigma^\ell \sqrt{\Delta\hat{T}^\ell} \ . \tag{12}$$

- *We use* $\widetilde{W}^j$ *to define a fixed grid as* ***relative strikes***

$$\hat{K} := \exp\left(z\widetilde{W}^1\right) \bigcup \exp\left(z^i \widetilde{W}^M\right) \ , \tag{13}$$

sufficiently sorted, with boundaries $\hat{K}^0 := \frac{1}{2}\hat{K}^1$ *and* $\hat{K}^{N+1} := \frac{3}{2}\hat{K}^N$*.*

- *Finally, we use the new real-valued ML-SANOS parameters to define the* ***squared discrete local volatilities****:*

$$\Sigma^{j,i} := \min\Big\{\Sigma_{\max}, \Sigma_{\min} + \exp(u^{\Sigma,j,i})\Big\} \ . \tag{14}$$

(This formulation avoid saturation of the sigmoid function as the raw log-DLV coordinate approaches the cap $\Sigma_{\max}$*; it requires adding a penalty term to the loss function.)*

Put together, this process describes a map

$$(\,u; z, \hat{T}\,) \overset{\substack{\mathrm{ML}\\ \mathrm{SANOS}}}{\longmapsto} (\,\Sigma, W;\, \hat{K}, \hat{T}) \overset{\mathrm{DLV}}{\longmapsto} (\,q, W;\, \hat{K}, \hat{T}) \overset{\mathrm{SANOS}}{\longmapsto} C \ . \tag{15}$$

In our new parametrization u *lives in an unrestricted space and is therefore particularly suited for ML/AI based learning. The expiry locations remain fixed.*

In our experiments we use the bounds $(\Sigma_{\min}, \Sigma_{\max}) = (0.01^2, 3^2)$, $(\sigma_{\min}, \sigma_{\max}) = (0.01, 1)$, and $(\eta_{\min}, \eta_{\max}) = (0.01, 0.50)$.

3.1 Learning ML-SANOS from real data

Our new parameterization allows the model to fit not only DLV's but also forward volatilities and, via u^η the size of the strike grid for a given set of market data. This extension means the model is no longer linear in its parameters. We therefore fit it with stochastic gradient descent methods.

Model specifications We apply this to two model specifications: one with weekly expiries only and one with weekly expiries and daily expiries in the front week. The purpose of this comparison is to assess whether a purely weekly model can capture front dynamics natively. The specific model expiries are:

$$\begin{aligned} \text{weekly:} \quad & (5, 10, 15, \ldots, 70)\text{i.e.} \quad M = 14 \text{ business days,} \\ \text{daily+weekly:} \quad & (1, 2, 3, 4, 5, 10, 15, \ldots, 70) \text{ i.e.} \quad M = 18 \text{ business days.} \end{aligned}$$

We use $N = 200$ strikes computed using (13) for z strikes within $(-3, 2)$. The native coordinates are:

$$u_t = \left(u_t^\Sigma,\, u_t^\sigma,\, u_t^\eta\right) \in \mathbb{R}^{M(N+1)+2}. \tag{16}$$

The weekly model therefore has 2,816 native coordinates while the the daily+weekly model has 3,620 native coordinates. For each geometry we learn a "direct" representation u_t for each historic date. We also learn an additional "affine-200" fit with representation

$$u_t = b + Bh_t, \qquad h_t \in \mathbb{R}^{200}, \tag{17}$$

in terms of learned $h_t \in \mathbb{R}^{200}$, and shared $b \in \mathbb{R}^{M(N+1)+2}$ and $B \in \mathbb{R}^{(M(N+1)+2)\times 200}$.[6]

Data and model geometries The empirical population sourced from WRDS' Option Metrics Ivy data base contains 891 S&P Index dates from 2022-01-03 through 2025-08-29. See appendix A for our baseline pipeline we use for all our research with this data set.

In addition to the general filters, we here also exclude options ourside the model's strike range (which is effectively $(-3, 2)$ NMS) and any options expiring past the last model expiry of 70 business days. We also exclude tomorrow's AM-settled options.

We retain 3,566,258 original call-or-put book observations for the daily+weekly model and 3,566,742 for the weekly model.[7] All options are represented in call-equivalent units through put–call parity; call and put books at the same expiry and strike remain separate observations and contribute separate loss terms.

Objective Function For quote ℓ on date t, let C_t^ℓ be the decoded option price, $M_t^\ell = (A_t^\ell + B_t^\ell)/2$, and $\gamma_t^\ell = (A_t^\ell - B_t^\ell)/2$. We fit in principle the same objective as in (4):

$$\mathcal{L}_t^{\text{quote}} = \frac{1}{L_t} \sum_{\ell=1}^{L_t} \frac{(C_t^\ell - A_t^\ell)^+ + (B_t^\ell - C_t^\ell)^+ + 0.05|C_t^\ell - M_t^\ell|}{(A_t^\ell - B_t^\ell)/2} \tag{18}$$

The native controls first fit an independent u_t on every date. The affine-200 model jointly updates b, B, the historical states h_t, and the stable weekly dynamics in Section 4. A gauge penalty controls the affine non-identifiability by keeping the fitted states at zero mean and unit population covariance across the historical dates.

[6] The choice of 200 is an empirical capacity compromise rather than a structural claim about the intrinsic dimension of option surfaces. It is large enough to preserve almost all of the direct native quote fit while remaining below the number of historical dates, so a full innovation covariance and stable joint state dynamics can still be estimated. We report the native-versus-200 fit loss explicitly below rather than treating this dimension as exact.

[7] The slight difference in retained observations arises from the difference in the effective strike range for each model.

Remark 3.2 (Full Objective)**.** *Two regularization penalties were used. First, write*

$$u_{\max}^{\Sigma} := \log(\Sigma_{\max} - \Sigma_{\min}), \qquad a := 0.10, \qquad s := 0.35, \tag{19}$$

and, for every raw log-DLV coordinate, define the smooth one-sided cap residual

$$r_t^{j,i} := \frac{a}{s} \log\left(1 + \exp\left(\frac{u_t^{\Sigma,j,i} - u_{\max}^{\Sigma}}{a}\right)\right). \tag{20}$$

With $D = 891$ historical dates, the full-data DLV-cap penalty corresponding to the equal-date minibatch estimator used in training was

$$\mathcal{L}_{\text{cap}} := \frac{1}{D}\sum_{t=1}^{D}\frac{1}{2}\sum_{j=1}^{M}\sum_{i=1}^{N}(r_t^{j,i})^2, \qquad \lambda_{\text{cap}} = 1. \tag{21}$$

Pricing itself used the mechanical cap in (14); this smooth penalty supplies a restoring gradient to the raw coordinate where the capped pricing map is flat. Far above the cap it approaches a quadratic penalty with log-scale standard deviation $s = 0.35$, while $a = 0.10$ controls the smooth transition around the cap.

Second, let $d = 200$ be the affine state dimension and define

$$\bar{h} := \frac{1}{D}\sum_{t=1}^{D} h_t, \qquad \widehat{\Gamma}_h := \frac{1}{D}\sum_{t=1}^{D}(h_t - \bar{h})(h_t - \bar{h})^{\top}. \tag{22}$$

The gauge penalty was

$$\mathcal{L}_{\text{gauge}} := \frac{1}{d}\|\bar{h}\|_2^2 + \frac{1}{d}\|\widehat{\Gamma}_h - I_d\|_F^2, \qquad \lambda_{\text{gauge}} = 0.10, \tag{23}$$

Thus the complete regularization penalty was

$$\mathcal{L}_{\text{penalty}} = \lambda_{\text{cap}}\mathcal{L}_{\text{cap}} + \lambda_{\text{gauge}}\mathcal{L}_{\text{gauge}}. \tag{24}$$

Operationally, training was alternating: the quote update included $\lambda_{\text{cap}}\mathcal{L}_{\text{cap}}$, whereas the dynamics update included $\lambda_{\text{gauge}}\mathcal{L}_{\text{gauge}}$. The covariance uses the population normalization $1/D$, rather than $1/(D-1)$. This gauge removes translations and non-orthogonal changes of scale and shear; orthogonal rotations and reflections remain immaterial.

3.1.1 Market fit

The primary measure of fit is the excess of the model price relative to the bid–ask spread: for fitted price C, bid B, and ask A, define excess in half-spread units by

$$e(C; B, A) := \frac{(C - A)^+ + (B - C)^+}{(A - B)/2}. \tag{25}$$

Thus $e = 0$ exactly when the model price lies inside bid–ask. We compute the percentage of options within bid/ask, and conditionally of being outside, the mean and median violation.

Table 1 gives the complete quote-population results for the fitted two-η models.

Model	Quotes	Inside bid–ask	Mean excess	Cond. mean	Cond. median
Weekly native	3,566,742	73.2678%	1.1101	4.1525	1.4261
Weekly affine-200	3,566,742	72.8728%	1.1017	4.0612	1.3925
Daily+weekly native	3,566,258	82.2809%	0.3120	1.7609	0.6920
Daily+weekly affine-200	3,566,258	**80.9211%**	0.3229	1.6924	0.6776

Table 1: Aggregate fit. Excess is defined by (25).

Affine compression costs only 0.39 percentage points for the weekly geometry and 1.36 percentage points for the daily+weekly geometry. By contrast, adding the four short-end nodes improves affine-200 fit by 8.05 percentage points. Expiry geometry, not latent-state capacity, is therefore the main bottleneck.

The mechanism is visible directly in table 2.

Model	1D	2D	3D	4D	5D
Weekly native	24.07%	32.47%	42.04%	55.66%	45.45%
Weekly affine-200	24.35%	32.65%	42.11%	55.26%	45.26%
Daily+weekly native	94.56%	95.21%	95.57%	95.77%	68.43%
Daily+weekly affine-200	88.59%	91.98%	93.68%	94.13%	67.58%

Table 2: Inside-spread rates at the five shortest nominal expiries.

The weekly model has no independent 1D–4D state directions and must extrapolate from 5D. The daily+weekly model represents front-end dislocations explicitly.

Figure 3 shows that this is a localized geometry effect, not a general loss of fit. The explicit daily nodes lift the 1D–4D inside-spread rate from roughly 24–55% to 89–94% in the affine model. Once the market expiry is comfortably beyond the first weekly node, the two geometries converge and both approach near-complete containment at the longer end. Within either geometry, the native and affine-200 curves remain close; compression is therefore secondary to where the decoder places its expiry nodes.

The same conclusion appears in the magnitude of misses. Conditional on a quote falling outside the spread, Figure 4 measures its distance in half-spread units. A weekly decoder forced to extrapolate before 5D can miss 1D and 2D quotes by more than ten half-spreads on average; explicit daily nodes reduce that short-end error to roughly one to three half-spreads. The gap narrows rapidly after the first week.

Figure 5 gives a cross-sectional view on 2024-06-14. At representative expiries of 10, 23, 43, and 67 business days, the independently fitted market SANOS surface and the completed weekly affine model track one another closely over the central liquid strike range. The longer-expiry panels also expose a material discrepancy at the far-left boundary; the native-versus-affine IV RMSE is 5.30 and 6.87 volatility points there, compared with 0.12 and 0.30 at 10 and 23 days. The example therefore supports the aggregate fit result while making clear that extrapolative wing coordinates remain a limitation.

Interpretation of the 80% number. The 80.92% headline is the affine-compression reconstruction test over the full 891-date population. After joint fitting of the stable state dynamics, the simulation checkpoint places 81.48% of all quotes inside bid–ask; its 2025 slice contains 707,630 observations and has an 80.56% inside-spread rate. The shared decoder and dynamics used all dates, so the latter is a chronological slice rather than a strict temporal holdout. We interpret it as evidence that the fit is not concentrated in the earlier sample, not as an untouched forecasting estimate.

4 Statistical option-surface state-space model

4.1 Weekly AR(1)

Let h_n be the affine SANOS state sampled at weekly intervals. The fitted statistical law is

$$h_{n+1} = m + A(h_n - m) + \varepsilon_{n+1}, \qquad \mathbb{E}_{\mathbb{P}}[\varepsilon_{n+1} \mid h_n] = 0, \qquad \mathrm{Cov}(\varepsilon_{n+1}) = Q. \tag{26}$$

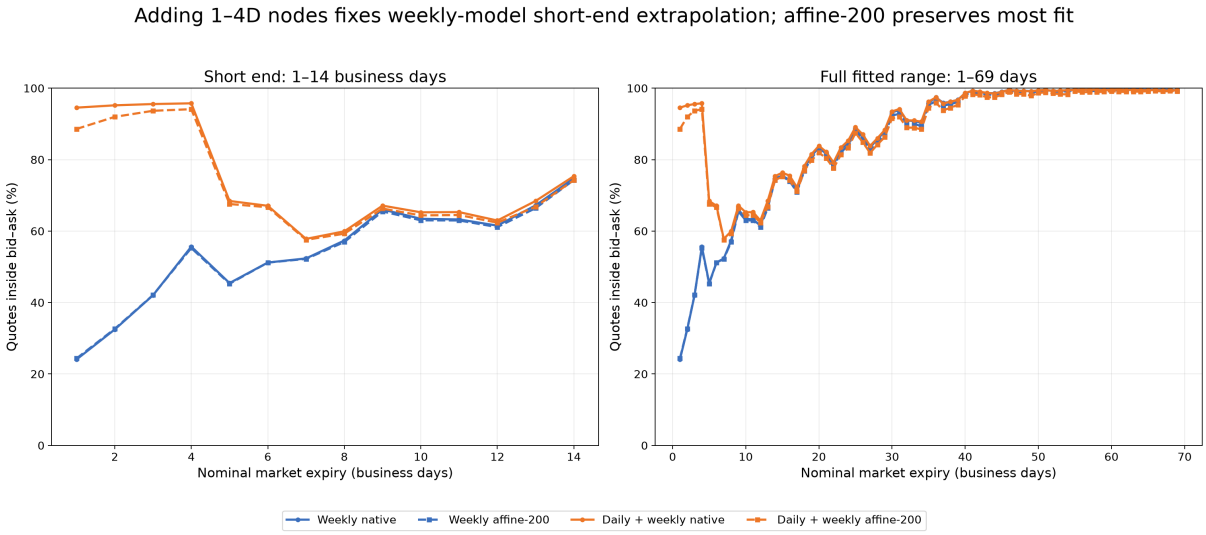


Figure 3: Inside-spread rate by nominal market expiry for the weekly and daily+weekly geometries, before and after affine-200 compression. All quote observations in each expiry bucket are included.

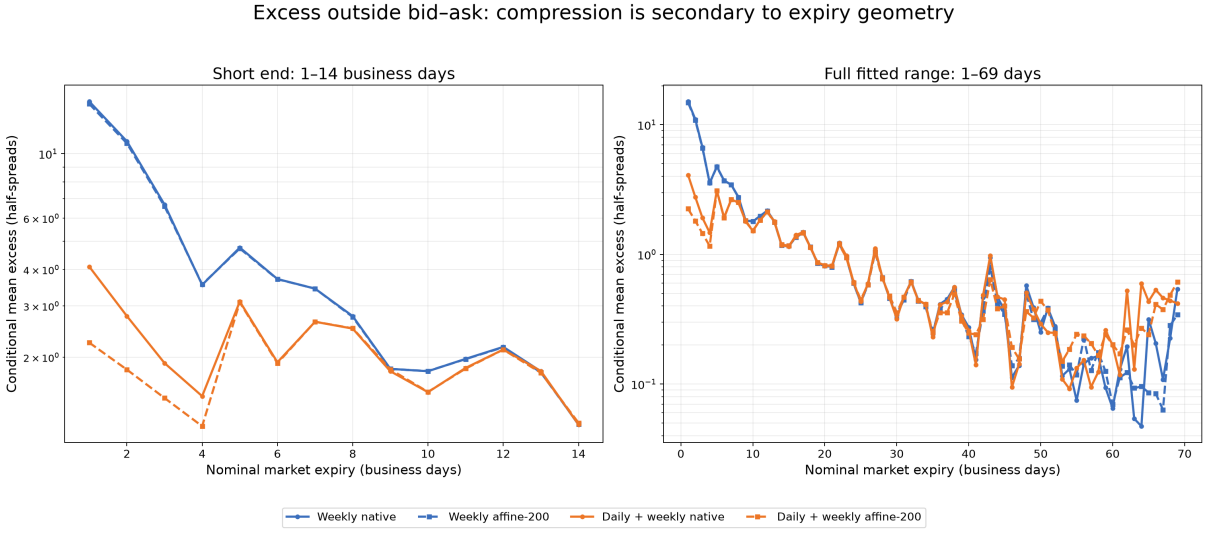


Figure 4: Mean half-spread excess by market expiry, conditional on the model price being outside bid–ask. The logarithmic vertical scale exposes the very large short-end extrapolation errors of the weekly geometry.

Rather than fitting an unrestricted discrete transition and checking it after the fact, the model parameterizes

$$\kappa = LL' + S + \kappa_{\min} I, \qquad S' = -S, \qquad A = e^{-\kappa}, \qquad Q = GG', \tag{27}$$

where L is lower triangular, S is unrestricted skew-symmetric, G is lower triangular with strictly positive diagonal, and $\kappa_{\min} = 10^{-3}$ per week. The symmetric part of κ is strictly positive, so every fitted A is stable throughout optimization. The state therefore has a finite invariant covariance P satisfying

$$P = APA' + Q. \tag{28}$$

In Gaussian mode we obtain exact fractional-week transitions. With $A_\Delta = e^{-\kappa\Delta}$,

$$h_{t+\Delta} = m + A_\Delta(h_t - m) + \varepsilon_\Delta, \qquad \varepsilon_\Delta \sim \mathcal{N}(0, Q_\Delta), \qquad Q_\Delta = P - A_\Delta P A'_\Delta. \tag{29}$$

This avoids Euler instability and permits daily, weekly, or longer Gaussian simulation steps from the same fitted weekly law.

4.2 Innovation laws and tail dynamics

Gaussian innovations are a convenient baseline but miss market tails. As a non-parametric comparator we also simulate at weekly steps by independently resampling centered fitted one-week residuals whose endpoints precede 2025. At two and three weeks, independently drawn residuals are propagated through A and summed; no multi-week block is replayed in this tail experiment.

For an exact-horizon state return r with fitted horizon covariance Q^H, define its normalized squared radius

$$(\delta^H)^2 := \frac{r'(Q^H)^{-1}r}{200}. \tag{30}$$

The Gaussian reference would concentrate this quantity sharply around one in 200 dimensions. The observed radii are both heavier-tailed and serially dependent.

4.2.1 No-replay stochastic-volatility diffusion

We therefore add a scalar latent log-variance diffusion without replaying any historical residual. Keeping the affine conditional mean and loading G from $Q = GG'$, the exact weekly transition is

$$\varepsilon_{n+1} = \sqrt{\rho \exp(g_n - v_g/2)}\, G z_{n+1}, \qquad z_{n+1} \sim \mathcal{N}(0, I), \tag{31}$$

$$g_{n+1} = \phi g_n + \sqrt{v_g(1-\phi^2)}\, \xi_{n+1}, \qquad \xi_{n+1} \sim \mathcal{N}(0, 1), \tag{32}$$

with independent z_{n+1} and ξ_{n+1}. Stationarity gives $g_n \sim \mathcal{N}(0, v_g)$ and hence $\mathbb{E}[\rho \exp(g_n - v_g/2)] = \rho$. Equation (32) is the exact discrete transition of a stationary Ornstein–Uhlenbeck diffusion; it therefore permits fractional-time simulation without Euler discretization.

The parameters (ρ, ϕ, v_g) are fitted on the 703 pre-2025 one-week innovations. Since

$$\log(\delta_n^2) = \log \rho + g_n - v_g/2 + \log(\chi^2_{200}/200), \tag{33}$$

we use a Kalman likelihood whose measurement mean and variance are the exact first two moments of $\log(\chi^2_{200}/200)$; only its Gaussian shape is an approximation. The weekly fit gives $(\rho, \phi, v_g) = (0.872, 0.249, 0.0556)$, and the daily+weekly fit gives $(0.878, 0.139, 0.0649)$. On the 158 2025 innovations, log-radius negative log likelihood falls from 1.986 to 0.089 in the weekly model and from 2.453 to 0.125 in the daily+weekly model relative to a pre-2025-scaled Gaussian benchmark.

For comparison we also fit variance-normalized multivariate Student-t innovations, either IID or with a persistent score-driven common scale. The fitted degrees of freedom are 42.6

and 34.5. On the same 2025 innovations the score-driven law reduces full-vector negative log likelihood per dimension from 1.377 to 1.364 in the weekly geometry and from 1.351 to 1.332 in the daily+weekly geometry; the improvement over IID Student-t is negligible.

Table 3 reports the probability of exceeding the pre-2025 empirical 95% radius at each horizon. "Market" is the realized 2025 rate; every simulated estimate uses one million paths.

Geometry	Innovation law	1W	2W	3W
Weekly	2025 market	5.063%	5.696%	5.696%
	Gaussian	0.1470%	0.00439%	0.00539%
	IID Student-t	4.350%	1.275%	1.164%
	Score-driven Student-t	4.523%	1.542%	1.435%
	Log-variance diffusion	3.960%	1.153%	1.032%
	Independent historical replay	5.143%	2.444%	2.386%
Daily+weekly	2025 market	6.329%	6.329%	5.696%
	Gaussian	0.0318%	0.000140%	0.000133%
	IID Student-t	4.139%	0.986%	0.856%
	Score-driven Student-t	4.119%	1.071%	0.931%
	Log-variance diffusion	3.751%	0.778%	0.659%
	Independent historical replay	5.150%	2.412%	2.318%

Table 3: Probability of exceeding the pre-2025 market 95% normalized-radius threshold. Market frequencies use 158 2025 transitions; simulation uses one million paths per law.

The no-replay diffusion is therefore a meaningful improvement, not a complete tail model. It also produces squared-radius serial correlations of 0.206 and 0.116, compared with 0.133 and 0.091 in the weekly and daily+weekly market histories; Gaussian and IID laws have zero population correlation. The 2025 sample is small and overlapping: 15-origin circular-block bootstrap 95% intervals for the market exceedance rates are approximately 0.6–14.6% across the six geometry/horizon cells. Differences of a percentage point should therefore not be over-interpreted. Independent historical replay is still closest in marginal tail mass, while the diffusion is a fully parametric generator with fresh support and explicit persistence.

4.3 Chronological surface-path comparison

Horizon radii test the latent innovation law but do not directly test paths of decoded option prices. We therefore form every available sequence of four fitted market-state surfaces separated by one full trading week, giving three consecutive weekly changes. Here "observed" means the date-specific latent state fitted to that day's irregular quote book and then passed through the common SANOS decoder; it is not a second interpolation of the raw quotes. Every candidate simulator starts from that fitted surface at the same origin date. Its path features are changes in normalized call prices on the Cartesian panel of target locations

$$T \in \{1, 5, 10, 20, 40, 70\} \text{ business days}, \qquad K/S \in \{0.85, 0.90, 0.95, 1.00, 1.05, 1.10, 1.15\}.$$

Each target expiry is mapped to the closest distinct model expiry. The daily+weekly geometry therefore has $3 \times 6 \times 7 = 126$ features; in the weekly geometry the 1D and 5D targets both map to 5D, leaving $3 \times 5 \times 7 = 105$. Coordinates are centered and scaled using pre-2025 observed paths only.

For this path-level test, the historical comparator replays an entire pre-2025 three-week sequence of fitted residuals, rather than drawing its three weekly residuals independently. It is therefore a deliberately strong finite-sample benchmark that preserves short-run clustering, but it is not a new parametric law and cannot generate innovations outside the stored blocks.

We report the standardized energy distance and an unbiased Gaussian-kernel maximum mean discrepancy (MMD) between observed and simulated 2025 paths. For the null distribution, entire

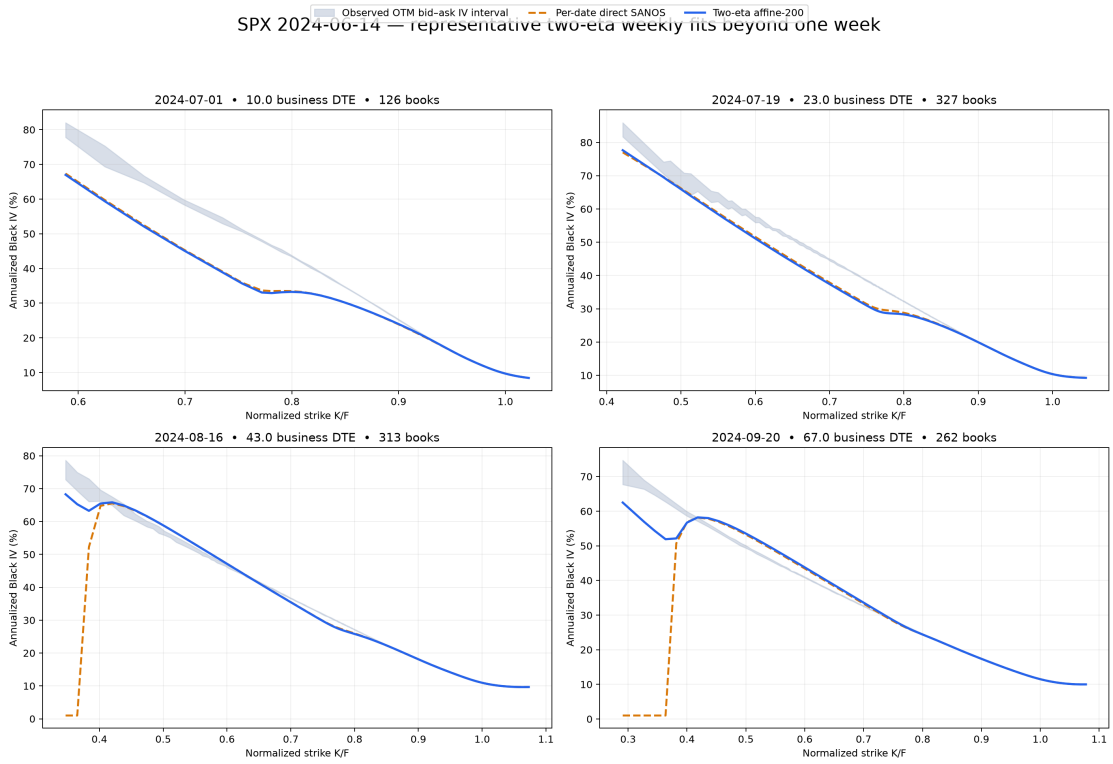


Figure 5: Representative SPX fit on 2024-06-14. The panels compare observed OTM bid–ask implied-volatility intervals, the per-date SANOS preprocessing fit, and the affine weekly model at four market expiries.

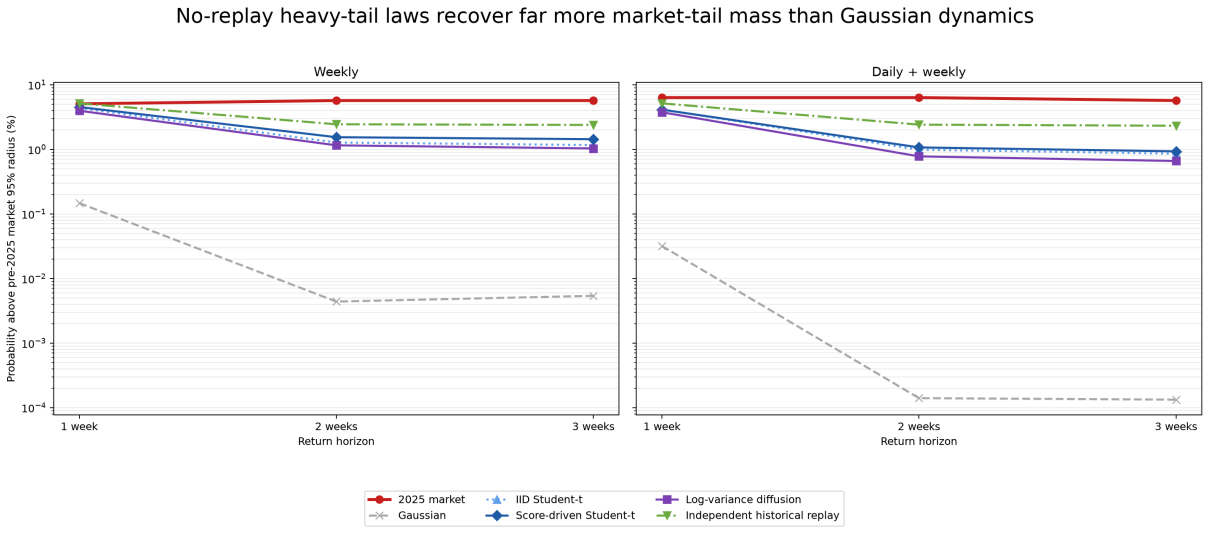


Figure 6: Probability of exceeding the pre-2025 market 95% normalized-radius threshold, on a logarithmic scale. Both no-replay heavy-tail laws recover orders of magnitude more tail mass than Gaussian innovations but remain too light at two and three weeks.

paired chronological blocks of 15 origin dates swap their observed/simulated labels; the block length matches the three-week feature horizon and retains most overlap-induced dependence. 1,999 random block swaps give a Monte Carlo p-value with minimum positive resolution 0.0005.

The 2025 split is a chronological holdout for fitting any auxiliary tail law, constructing a historical-residual pool, and estimating the feature standardization. The shared affine decoder and the baseline Gaussian AR law are nevertheless taken from the all-history checkpoint. The path comparison is therefore a holdout diagnostic for the innovation-law extensions conditional on the learned representation, not a fully frozen pre-2025 model forecast.

Table 4 reports 144 paired 2025 paths for each geometry. The p-value is the block-swap test for MMD; energy distance is reported as a second scale-sensitive discrepancy. None of the five candidates is rejected at 5%. This is evidence against a gross path-distribution failure, but not a ranking theorem: IID Student-t has the smallest weekly energy distance, whereas Gaussian has the smallest daily+weekly energy and MMD. The log-variance diffusion remains statistically plausible while improving the separate tail and persistence diagnostics above.

	Weekly			Daily+weekly		
Innovation law	Energy	$10^3\times\text{MMD}^2$	p	Energy	$10^3\times\text{MMD}^2$	p
Gaussian	0.572	4.602	0.550	0.810	1.594	0.704
IID Student-t	0.540	5.016	0.344	0.854	3.788	0.457
Score-driven Student-t	0.569	4.389	0.225	0.838	3.786	0.416
Log-variance diffusion	0.606	5.476	0.378	0.854	3.480	0.415
Historical block replay	0.592	6.671	0.328	0.857	6.968	0.345

Table 4: Chronological decoded-surface path diagnostics on 144 paired 2025 origins. Each path contains three consecutive weekly changes; p is based on 1,999 paired 15-origin block swaps.

4.4 Stability diagnostics

Statistic	Weekly	Daily+weekly
State dimension	200	200
Dates	891	891
Exact 1W innovations	861	861
$\rho(A)$	0.9577	0.9561
Slowest-mode half-life	16.1 weeks	15.4 weeks
Minimum real eigenvalue of κ	0.0432	0.0449
Invariant covariance condition number	3.51	3.64
Invariant covariance effective rank	196.5	196.4

Table 5: Fitted state-law diagnostics.

Both laws are stable and retain almost full effective rank in state space. A simulator can therefore start from an observed state or draw $h_0 \sim \mathcal{N}(m, P)$ when a stationary initialization is appropriate.

4.5 Conditional front-expiry spot block

Remark 4.1 (Normalization with forwards and discount factors)**.** *If the market has discount factors* $\boldsymbol{DF}_t(T) = \exp(-R_t(T))$ *and forwards* $\boldsymbol{F}_t(T) = \boldsymbol{S}_t \exp(R_t(T) - D_t(T))$*, the normalized one-step log-spot return is*

$$\Delta s_{t+1} = \log \boldsymbol{S}_{t+1} - \log\big(\boldsymbol{DF}_t(\Delta t)\boldsymbol{F}_t(\Delta t)\big) = \log \boldsymbol{S}_{t+1} - \log \boldsymbol{S}_t + D_t(\Delta t).$$

Option prices require no further drift adjustment here because the input surface has already been expressed in forward- and discount-normalized units.

The normalized option surface can be simulated autonomously. Cash spot is attached through the front-expiry SANOS distribution while preserving its observed contemporaneous dependence with surface innovations. Let F_t^1 denote the CDF obtained from the first decoded SANOS density, including the lognormal backbone, and let z_{t+1} be the whitened one-session state innovation. Each observed gross spot return is transformed to its front-law Gaussian score

$$g_{t+1} = \Phi^{-1}\left[F_t^1\left(\frac{S_{t+1}}{S_t}\right)\right]. \tag{34}$$

On transitions preceding the evaluation window, we estimate a ridge loading b by regressing the standardized g_{t+1} on z_{t+1}. Its penalty is selected by time-ordered validation. If $\sigma_\perp$ is the fitted residual standard deviation, a joint scenario uses independent standard normals $Z \in \mathbb{R}^{200}$ and $\epsilon \in \mathbb{R}$ and sets

$$G = \frac{b'Z + \sigma_\perp \epsilon}{\sqrt{b'b + \sigma_\perp^2}}, \qquad \frac{S_{t+1}}{S_t} = (F_t^1)^{-1}(\Phi(G)). \tag{35}$$

The same Z drives the next surface state. Consequently the spot draw has the current front-expiry marginal and the fitted spot–surface copula rather than being independent of the surface shock. For each date we tabulate F_t^1 on a fine log-return grid and invert it with a monotone PCHIP spline. This is an interpolation of the quantile function; it avoids a separate root solve for every scenario without sacrificing monotonicity. Log-spot itself is not part of the invariant surface state.

Because F_t^1 is the option-implied risk-neutral front-expiry marginal, this construction does not estimate or add a physical spot drift. Its first moment is fixed by the observed front forward (and equals one in forward-normalized units), so spot carries the one-step risk-neutral drift implied by market carry. The fitted copula changes only the dependence between spot and surface innovations; it does not change the spot marginal or its drift.

4.6 Decoded implied-volatility geometry

Although h_t has 200 coordinates, decoded implied-volatility variation is much lower-dimensional. We differentiate the affine SANOS decoder at the invariant mean and combine its Jacobian with P. The resulting PCs are directions in decoded implied-volatility space, not the learned state coordinates.

Cumulative decoded-IV variance	Weekly	Daily+weekly
PC1	74.85%	77.68%
PC1–2	86.97%	86.98%
PC1–4	95.63%	94.50%
PC1–8	99.46%	99.39%

Table 6: Decoded implied-volatility variance under the invariant state law.

Figure 7 gives the full leading spectrum. PC1 is a nearly parallel volatility-level move and explains 74.85% of weekly decoded variance and 77.68% of daily+weekly decoded variance. Four decoded-IV factors explain 95.63% and 94.50%, respectively. This does not imply that four latent coordinates would fit the quote book: the PCA is performed *after* the 200-state model has been passed through the nonlinear SANOS decoder and weighted by its invariant covariance.

The invariant means in Figure 8 are the natural "base volatility" surfaces of the two statistical models. Their broad term-structure and skew geometry is similar. The difference is concentrated

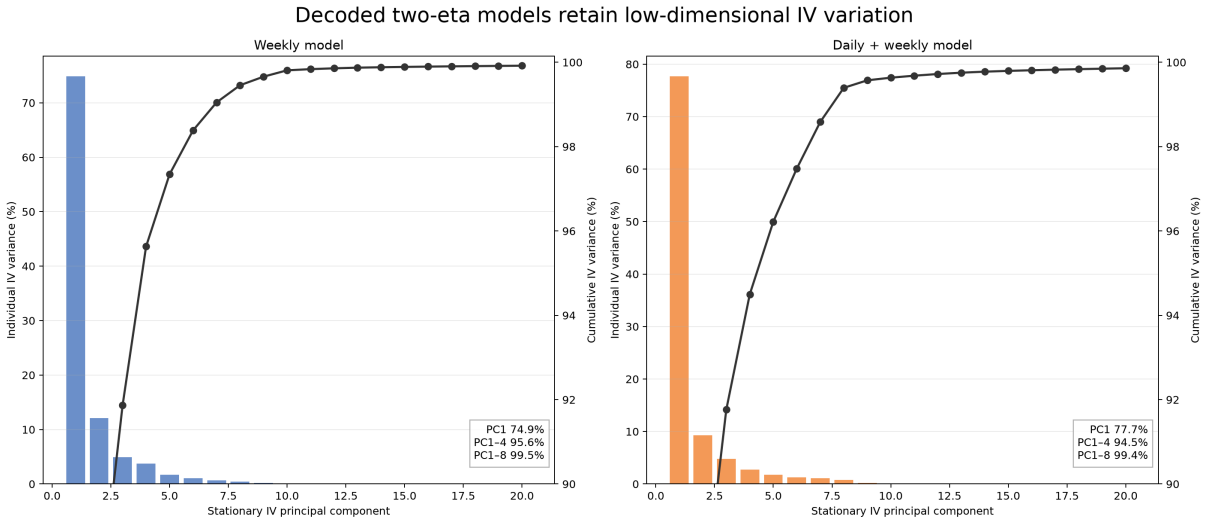


Figure 7: Variance explained by PCA of decoded implied-volatility levels under the fitted invariant state laws. The 200 affine coordinates are not themselves these PCs.

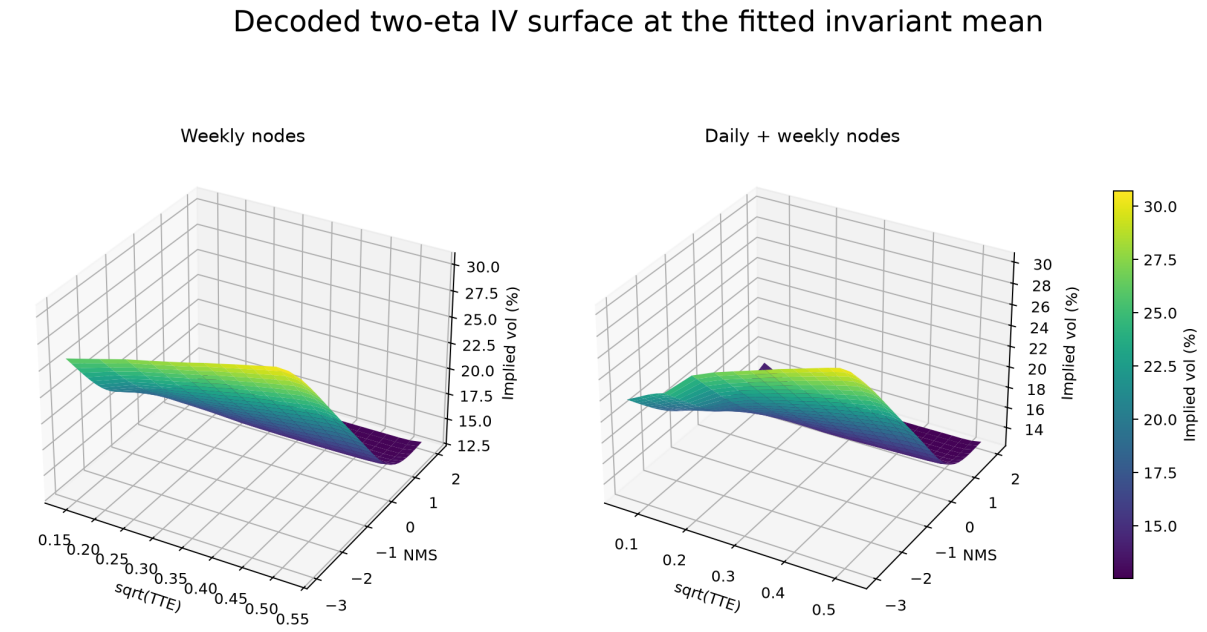


Figure 8: Base implied-volatility surfaces, obtained by evaluating each affine SANOS decoder at the invariant state mean. Axes are square-root time-to-expiry, normalized moneyness, and annualized implied volatility.

at the front: the daily+weekly model has four additional expiry slices with independent DLV and forward-volatility coordinates, whereas the weekly model extrapolates from 5D.

Figures 9 and 10 show the first eight eigensurfaces. Each panel is one invariant standard deviation and has its own vertical scale; PCA signs are arbitrary. PC1 is the global level. PC2 is primarily a broad skew/term-structure rotation, while PC3 couples skew and maturity. Higher modes become increasingly localized. The daily+weekly model places visibly more of those localized modes at 1D–4D, which is precisely the flexibility needed to fit the front-end quote book.

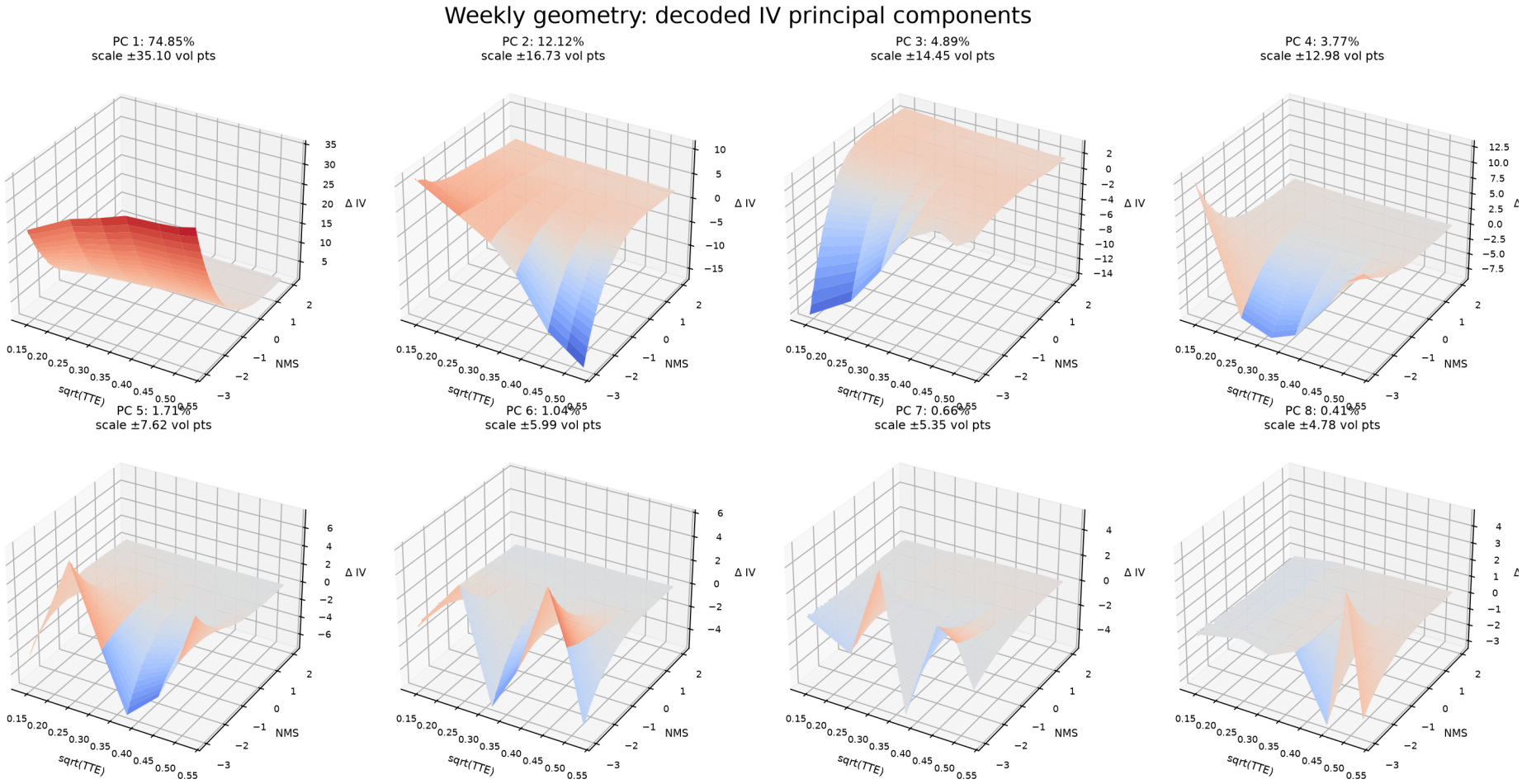


Figure 9: Weekly geometry: decoded implied-volatility PCs 1–8 at one invariant standard deviation. The title of each panel reports explained variance and the panel-specific move in volatility points.

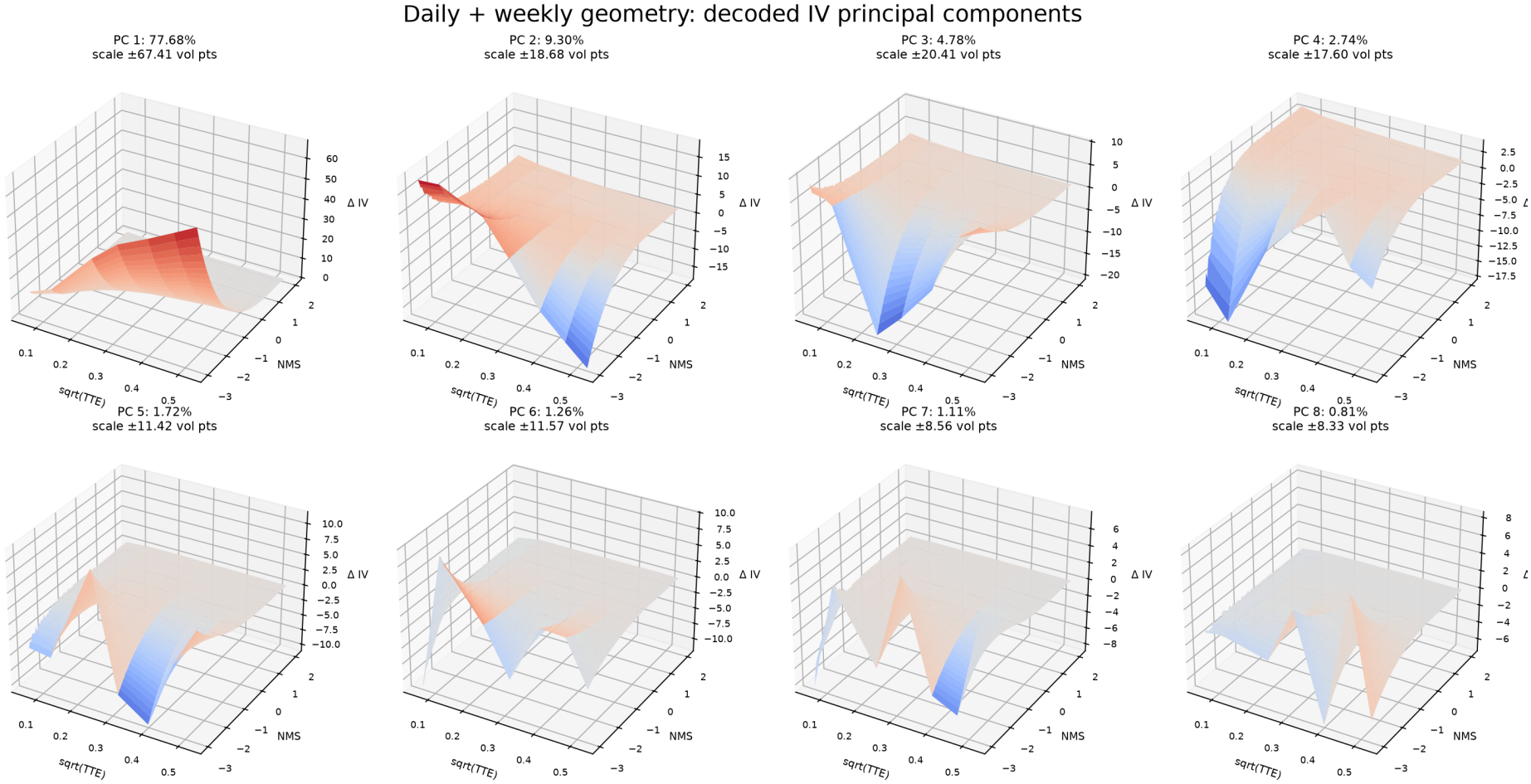


Figure 10: Daily+weekly geometry: decoded implied-volatility PCs 1–8 at one invariant standard deviation. Additional front-end modes appear because 1D–4D are explicit model expiries.

4.6.1 Invariant-distribution visualization

The PCA summarizes local covariance at the invariant mean; complete nonlinear decodes verify what that covariance produces as a surface distribution. Figure 11 shows six independent draws from each fitted $\mathcal{N}(m, P)$ state law. The weekly model's sub-5D ATM term structure is visibly constrained by extrapolation from its first node. The daily+weekly model can generate sharp but still statically admissible front-end dislocations.

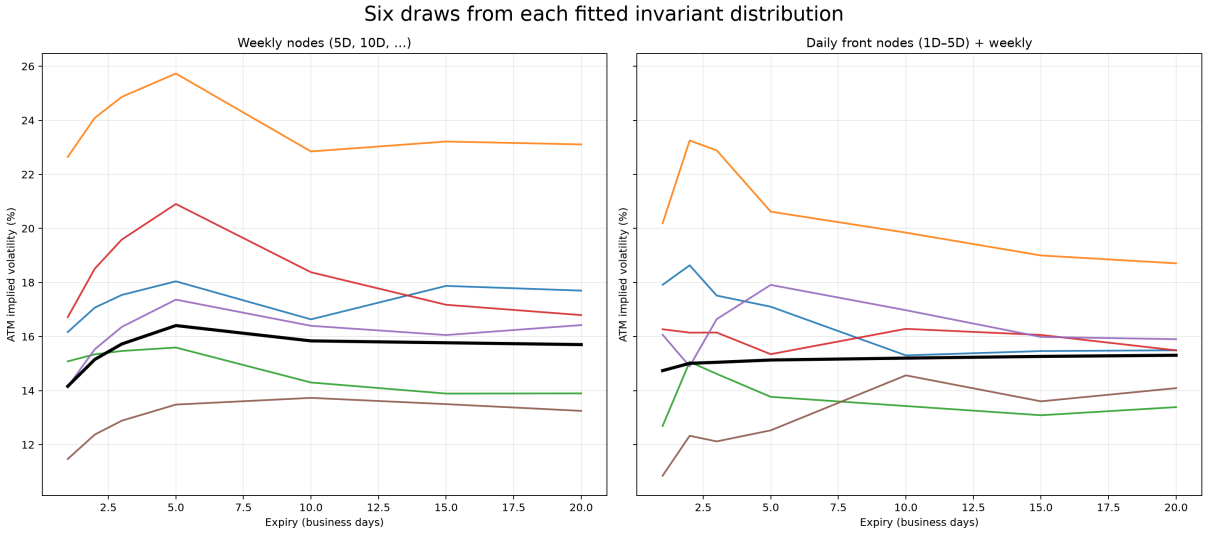


Figure 11: ATM implied-volatility term structures from six independent invariant-state draws for each model. Black is the invariant-mean decode. These are stationary cross-sectional samples, not consecutive time steps.

Figure 12 uses the same 200-dimensional standard-normal seed for both models, mapped through each model's own invariant covariance and decoder. It provides a controlled visual comparison of complete draws: their long-end shapes remain broadly comparable, while the daily+weekly surface has materially richer short-end geometry.

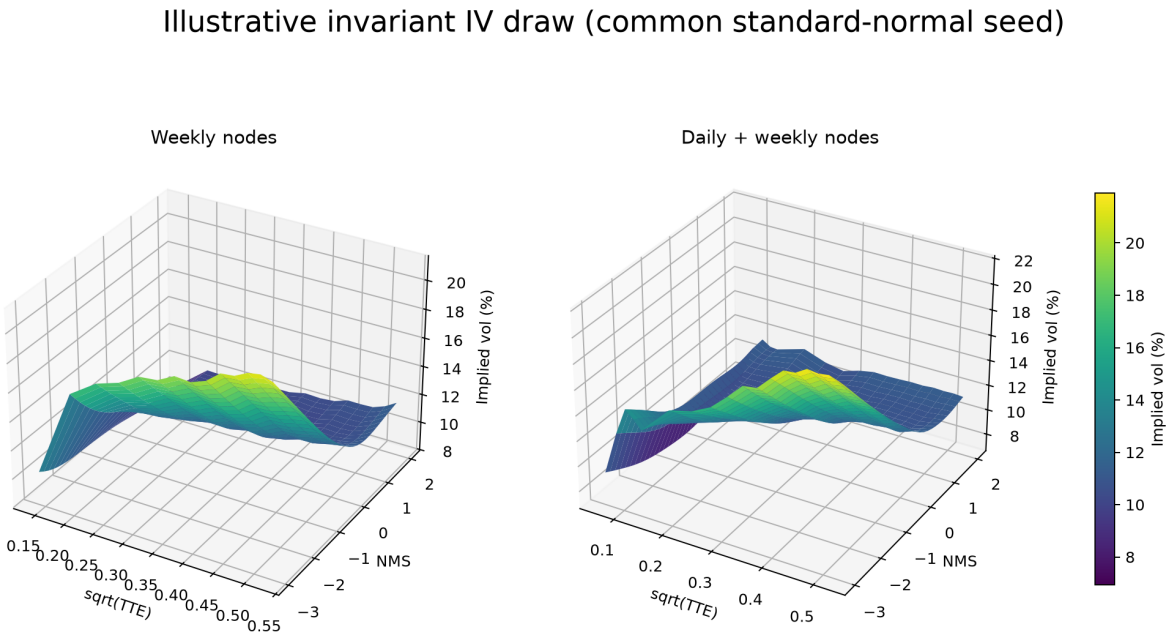


Figure 12: Paired complete invariant-state surface draws. A common standard normal seed is transformed by each fitted invariant covariance before being decoded by the corresponding affine SANOS model.

4.7 Simulation algorithm

The statistical DYSANOS simulator is summarized as follows.

1. Choose the daily+weekly geometry and initialize h_0 from an observed date or from the invariant state distribution.
2. Propagate the state with the exact Gaussian transition (29), or use the no-replay log-variance diffusion (31)–(32) when persistent tails are required. Historical innovation replay remains a non-parametric weekly benchmark.
3. If cash prices are required, propagate s_t conditionally using (35).
4. Form $u_t = b + Bh_t$, map it to bounded SANOS parameters through (14) and (12), and apply the density recursion of Section 3.
5. Evaluate $C_t(T, K)$ at any desired strike and expiry. For a fixed cash contract (τ, k) use $T = \tau - t$ and $K = k/S_t$.

The AR transition and SANOS decoder serve different roles. Stability of the first supplies a coherent long-run statistical state process; the second guarantees a valid option cross-section for every realized state. The combination is what we mean by consistent $\mathbb{P}$-simulation in this paper.

4.8 Fixed-sample to-expiry entropy projection

As a finite-sample diagnostic, we ask whether the simulated law can be reweighted so that a rolling option book has no expected return outside an explicit transaction-cost band. Both experiments use one historical initial state (25 October 2023) and $N = 1{,}000{,}000$ antithetic simulated paths. Each option has a fixed cash strike set on its entry date and is held to its terminal payoff. The admissible expected-return band is 1bp for spot and 10bp for each option, representing symmetric entry costs.

We considered two nested books. The pilot book contains spot and, for each of 5D, 10D, 15D, and 20D, a call struck at $K/S_t = 0.99$, an ATM call, and a put struck at $K/S_t = 1.01$. Repeating these 13 instruments on each of 25 entry dates gives $25 \times 13 = 325$ return coordinates.

The expanded book replaces those three options by 11 normalized-moneyness coordinates at every expiry,

$$z \in \{-1.5, -1.3, -1.1, -0.9, -0.7, -0.5, -0.3, -0.1, 0.1, 0.3, 0.5\}. \tag{36}$$

At entry time t, normalized moneyness z is converted to the cash strike

$$K = S_t \exp\big(z\,\nu_t(T)\big), \qquad \nu_t(T) = \sqrt{\frac{\sigma_{t,\epsilon}^2\epsilon + \widetilde{W}_t(T)^2}{1+\epsilon}}, \tag{37}$$

where $\widetilde{W}_t(T)$ is the strike-grid total volatility from (12), $\sigma_{t,\epsilon} = \widetilde{W}_t(\hat{T}^1)/\sqrt{\hat{T}^1}$, and ϵ is the canonical small-time regularization. Negative-z contracts are ITM calls and positive-z contracts are ITM puts. The grid in (36) deliberately does not contain zero, so this expanded book has no ATM option. With spot and 4×11 options at each entry date it has $25 \times 45 = 1{,}125$ return coordinates.

Let $R_n \in \mathbb{R}^d$ be the return vector on path n, where $d \in \{325, 1{,}125\}$, let $c \in \mathbb{R}^d_+$ be the cost band, and let w_n denote an empirical density relative to the uniformly weighted simulation

sample. We solve the convex information projection

$$\begin{aligned} \min_{w_1,\dots,w_N>0} \quad & \frac{1}{N}\sum_{n=1}^{N} w_n \log w_n, \\ \text{subject to} \quad & \frac{1}{N}\sum_{n=1}^{N} w_n = 1, \\ & -c \;\le\; \frac{1}{N}\sum_{n=1}^{N} w_n R_n \;\le\; c. \end{aligned} \tag{38}$$

The dual is a signed entropic OCE objective. We solve it with an active-set method and streamed log-sum-exp evaluations, so the complete million-path return matrix need not be copied into the optimizer.

Diagnostic	Three-option pilot	11-option NMS book
Paths	1,000,000	1,000,000
Options per expiry	3	11
Return coordinates	325	1,125
Active coordinates	207	336
Relative entropy $N^{-1}\sum_n w_n \log w_n$	0.0619141	0.1130914
Effective sample size	877,973 (87.80%)	765,493 (76.55%)
Minimum density weight	0.00880	0.00286
Maximum density weight	36.42	84.17
Zero density weights	0	0
Maximum cost-band violation	8.45×10^{-8}	3.15×10^{-8}
Maximum KKT residual	8.45×10^{-8}	3.53×10^{-8}
Primal–dual gap	7.14×10^{-9}	3.37×10^{-9}

Table 7: In-sample convex entropy projection for rolling to-expiry returns. Residuals are in return units. The maximum cost-band violations are 0.000845bp and 0.000315bp, respectively.

All fitted weights are strictly positive. The expanded book retains 76.55% of the original effective sample size despite imposing 1,125 moment bands. For both books the optimizer terminated during its final line search rather than returning its strict success flag. The independently recomputed primal, KKT, and primal–dual residuals in Table 7 show that the economically relevant cost-band constraints are nevertheless satisfied to substantially less than one thousandth of a basis point.

This is deliberately an *in-sample, unconditional* statement. The weights solve empirical moment constraints for this initial state and this fixed path population. They do not prove the existence of a single state-conditional transition density, do not constitute an out-of-sample forecast, and do not alter the statistical dynamics used elsewhere in the paper. The experiment nevertheless establishes that the selected finite book is compatible with a well-conditioned, absolutely continuous empirical reweighting when contracts are evaluated to expiry.

Section 5 builds on this reweighting with a genuinely state-conditional construction.

4.9 Scope and limitations

The 80.92% market-fit figure is computed on the population used to estimate the final all-history model. It is not a claim of 80% future forecast coverage. A formal forecasting study should freeze the decoder and dynamics before the evaluation period.

The Gaussian state law severely understates tail frequency. The no-replay log-variance diffusion improves likelihood, tail mass, and serial-tail dependence, but still underproduces two- and three-week extremes. Historical innovation replay is closer in marginal tail mass, while the

path-level tests have only 144 overlapping 2025 origins and do not distinguish the candidates sharply. A richer multiscale or regime-dependent volatility law is the most direct next extension.

Finally, static arbitrage-freedom is a property of each decoded surface. It does not by itself impose restrictions on expected returns of fixed contracts through simulated time. The fixed-sample entropy projection in Section 4.8 is an unconditional diagnostic for one finite simulated population, not an additional dynamic pricing assertion. Section 5 reports a first, explicitly reduced attempt at a genuine dynamic construction.

5 Risk-Neutral DYSANOS

Static arbitrage-freedom of each decoded surface does not by itself impose a martingale restriction on the returns of fixed contracts through simulated time (Section 4.8). We here report a first, deliberately reduced demonstration that a genuine *near-martingale* measure can nevertheless be constructed for a subset of tradable contracts by applying the Deep Hedging drift-removal method of [BMPW22a, BMPW22b] directly to simulated DYSANOS paths. We do not claim risk-neutrality of the full surface; extending the construction to *all* options is ongoing work.

5.1 Universe and pre-conditioning

We trade spot and, at each of 25 fixed trading dates, floating options at the four expiries $5D, 10D, 15D, 20D$ struck on the same 11-point normalized-moneyness grid (36) via (37); once entered, each option is a fixed contract (τ, k) held to its own expiry. This is the same $44+1=45$-dimensional "expanded book" and the same $N = 1{,}000{,}000$ fixed simulated paths used for the 11-option NMS entropy projection in Table 7. Writing C_t^i for the entry price and V_t^i for the (spot or option) expiry value of instrument i, returns are expressed as fractional gains,

$$R_t^i = \frac{V_t^i}{C_t^i} - 1, \qquad i = 1, \dots, 45, \tag{39}$$

and entry costs are $c^{\text{spot}} = 1\text{bp}$ and $c^{\text{opt}} = 10\text{bp}$ of traded dollar notional.[8]

As a pre-condition we reuse the convex projection (38): its solution w defines a reweighted measure $\mathbb{P}^w$ via $d\mathbb{P}^w/d\mathbb{P}|_{\{\omega_n\}} = w_n$, under which every return in the book has zero unconditional drift up to its cost band, $\left|\mathbb{E}^{\mathbb{P}^w}[R_t^i]\right| \le c^i$.

5.2 A near-martingale measure via Deep Hedging

Let u be a smooth, strictly increasing, concave utility with $u(0) = 0$, $u'(0) = 1$, $\lim_{x\to-\infty} u'(x) = \infty$, $\lim_{x\to\infty} u'(x) = 0$; our experiments use the entropy utility $u(x) = (1 - e^{-\lambda x})/\lambda$. Following [BTT07], the associated OCE monetary utility

$$U(X) := \sup_{y\in\mathbb{R}} \mathbb{E}^w\big[u(y + X) - y\big] \tag{40}$$

is increasing, concave, and cash-invariant, $U(X + c) = U(X) + c$ for $c \in \mathbb{R}$; for the entropy utility it reduces to the entropic certainty equivalent $U(X) = -\lambda^{-1} \log \mathbb{E}^w[\exp(-\lambda X)]$.

Trading actions are parameterized directly on the state $x_t = (s_t, h_t)$ already used for $\mathbb{P}$-simulation (Section 4.7),

$$a_\theta(x_t) = W_\theta x_t + F_\theta(x_t) + b_\theta \ \in \mathbb{R}^{45}, \tag{41}$$

where W_θ is a matrix, b_θ a vector, and F_θ a three-layer SeLU network with 128 inner nodes per layer.

[8] For numerical stability the Deep Hedging training in this section uses one tenth of these costs.

We solve the empty-portfolio Deep Hedging problem under $\mathbb{P}^w$,

$$\sup_{\theta,\, y\in\mathbb{R}} \mathbb{E}^{\mathbb{P}^w}\Big[u\Big(y + \textstyle\sum_t a_\theta(x_t)' R_t - |a_\theta(x_t)|' c\Big) - y\Big] \;\longrightarrow\; \theta^*, y^*, \tag{42}$$

and define the candidate density

$$\frac{d\mathbb{Q}}{d\mathbb{P}^w} := u'\Big(y^* + \textstyle\sum_t a_{\theta^*}(x_t)' R_t - |a_{\theta^*}(x_t)|' c\Big). \tag{43}$$

By the theorem of [BMPW22b], for a ranging over all measurable strategies $\mathbb{Q}$ is a *near-martingale measure*: every conditional expected return is bounded by its own transaction cost, $|\mathbb{E}_t^{\mathbb{Q}}[R_t^i]| \le c^i$. For the network class (41) the statement holds within that strategy class rather than over all measurable strategies.

5.3 Empirical confirmation

If $\mathbb{Q}$ is indeed a near-martingale measure, a Deep Hedging agent solving the same empty-portfolio problem under $\mathbb{Q}$, within the same action class but with an independently re-optimized $\tilde\theta, \tilde y$,

$$U_{\mathbb{Q}}^*(0) := \sup_{\tilde\theta,\, \tilde y\in\mathbb{R}} \mathbb{E}^{\mathbb{Q}}\Big[u\Big(\tilde y + \textstyle\sum_t a_{\tilde\theta}(x_t)' R_t - |a_{\tilde\theta}(x_t)|' c\Big) - \tilde y\Big], \tag{44}$$

should be unable to extract a positive objective value. For our reduced 45-instrument, 25-date, $N = 1{,}000{,}000$-path universe we obtain $U_{\mathbb{Q}}^*(0) = -0.000695 \pm 0.000414$ at the 95% level, statistically indistinguishable from zero and consistent with $\mathbb{Q}$ being a near-martingale measure for this book.

5.4 Scope and limitations

This is a reduced, "V1.0" demonstration rather than a general construction. The experiment is numerically heavy (roughly 10 hours for $N = 1{,}000{,}000$ paths and 25 trading dates) and, despite the theoretical guarantee, not very stable numerically; both runs use full-batch training on the complete simulated set with no held-out validation split. The resulting $\mathbb{Q}$ is a near-martingale measure for the stated 45-instrument book and strategy class only; it provides no risk-neutrality statement for other strikes, expiries, or trading frequencies, and the model-free reweighting step relies on having many more simulated paths than tradable instruments. Reducing the method to a form that achieves risk-neutrality for the *full* DYSANOS surface is the natural next step.

6 Conclusion

We presented with DYSANOS the first generative option market model built from smooth, strictly statically arbitrage-free SANOS surfaces. Our approach does not rely on approximate static constraints; each smooth option surface is strictly arbitrage free and allows pricing of any option on GPU. We have shown how to leverage our representation to build a simple, but already fairly powerful baseline model which can generate spot and option paths for many days in the future up to years. The surface-first AR(1) captures first-order level, skew, term, and leverage effects. Its front and term smoothing coordinates allow the short end to move separately from later expiries, and the daily+weekly affine model places 80.92% of the full quote population inside bid–ask. A no-replay log-variance diffusion improves the deficient Gaussian tail law and reproduces tail persistence, although multi-week tail mass remains too low. The decoded-surface path diagnostics do not reject any of the Gaussian, heavy-tail, diffusion, or replay candidates at the 5% level, although their power is limited by the 144 available overlapping origins. For one million simulated paths from a representative initial state, a convex entropy projection

of 1,125 rolling to-expiry return coordinates remains well-conditioned (76.55% ESS), retains strictly positive weights, and satisfies the stated transaction-cost bands to less than 0.001bp in sample. For a reduced 45-instrument book on the same simulated paths, a Deep Hedging construction (Section 5) goes further and builds a state-conditional near-martingale measure $\mathbb{Q}$: an independently re-optimized hedging agent cannot extract a positive objective value under $\mathbb{Q}$ ($U^*_{\mathbb{Q}}(0) = -0.000695 \pm 0.000414$ at the 95% level). This is a reduced, computationally heavy first demonstration rather than a full-surface risk-neutral construction, which remains future work.

We have provided a baseline model to generate long paths of daily spot and option prices. Each option surface is free of static arbitrage. Moreover, we can price *any* option on each path given the ML-SANOS parameters for the respective sample and time step. The resulting paths therefore provide statistically consistent prices for arbitrary fixed cash contracts rather than only for a preselected finite set.

7 Thanks

We want to thank the team at Wharton for their WRDS https://wrds-www.wharton.upenn.edu portal and Option Metrics https://optionmetrics.com/united-states for providing their IvyDB data via this service. Without this data our work would not have been possible.

A Appendix: Data Pipeline

In the interest of full transparency we here discuss main elements of the data pipeline used in our experiments.

The market-data pipeline constructs a standardized daily time series of SPX option surfaces. Raw option quotes are cleaned, normalized, filtered, checked against an arbitrage-free reference surface, and finally collected into irregular quote books suitable for direct model training.

The pipeline processes SPX option data between January 2nd, 2020 and August 29th, 2025. The latter is the latest day available at the time of writing. Only valid full-day NYSE trading days are considered.

A.1 Pipeline Overview

The complete transformation can be summarized as

$$\begin{aligned}\text{raw SPX option quotes} &\longrightarrow \text{daily cleaning and enrichment}\\ &\longrightarrow \text{arbitrage-free quote surface using linear SANOS}\\ &\longrightarrow \text{liquidity and moneyness filters}\\ &\longrightarrow \text{shared 200-strike SANOS model geometry}\\ &\longrightarrow \text{direct native and affine-200 quote fits.}\end{aligned}$$

The key insight when working with Option Metrics IvyDB data is that the forward curves from the `Forward_Price` table do not line up with the option data themselves. We therefore compute our own forward from weighted put-call parities. We also convert the tightest options into call prices. Such data still has arbitrage, hence we use a linear SANOS pre-processing step to obtain fitted prices within bid/ask which adhere to basic linear no-arbitrage conditions (the absence of which would indicate actionable arbitrage). We drop any strikes where such a fit cannot be achieved, assuming that fit issues are due to data alignment challenges rather than genuine trading opportunities.

A.2 Raw Daily Market Data

For every eligible date d, the underlying data contain quantities such as

- the SPX spot price S_d and previous spot price S_{d-1};
- option strikes and expiration dates; call and put bid and ask quotes; volume and open interest; AM/PM settlement indicators;
- forward and discount-curve information (with forwards being re-calculated below).

The default raw-data cleaning removes

- same-day expirations;
- options with bids below 0.011 (the minimum tick size is 0.05$ for series trading below 3$);
- observations with missing or invalid quotes; observations with an invalid settlement flag; irregular expirations; dates or ticker-date combinations known to contain bad data (e.g. 2025-01-08 which was a holiday).

A.3 Enrichment and Expiry Filtering

The cleaned quotes are enriched with

- business days and year fractions to expiry where AM options are counted as 20% of the day;
- implied forward prices computed off put-call parity per expiry for all strikes for which both call and put were traded and have valid bid/ask prices and sizes, weighted by put-call implied forward bid-ask spread.[9]
- using the new forward, define "market" mid call prices between the actual call prices and synthetic call prices from the puts;
- compute implied volatilities; forward- and discount-adjusted prices, strikes, and Greeks.

An expiration is retained only if it satisfies the default liquidity and quality requirements. In particular,

- at least 10 strikes must have either a traded call or put;
- at least 4 strikes must have both a traded call and put;
- the discounted forward normalized spread must be at least 10^{-6};
- implied volatility must lie in

$$0.01 \leq \sigma^{\text{impl}} \leq 5.$$

[9] We compute $\text{fwd}^{\text{ask}}(K) := +\text{Call}^{\text{ask}} - \text{Put}^{\text{bid}} - \text{df}K$ and $\text{fwd}^{\text{bid}}(K) := +\text{Call}^{\text{bid}} - \text{Put}^{\text{ask}} - \text{df}K$ using the discount factor sourced from IvyDB. Then we define $\text{fwd}^{\text{mid}}(K) := 0.5\,(\text{fwd}^{\text{ask}}(K) + \text{fwd}^{\text{bid}}(K))$ and $\gamma^{\text{mid}}(K) := 0.5\,(\text{fwd}^{\text{ask}}(K) - \text{fwd}^{\text{bid}}(K))$ and

$$\text{fwd} := \frac{\sum_K \gamma(K)^{-1}\text{fwd}^{\text{mid}}(K)}{\sum_K \gamma(K)^{-1}}$$

A.4 Arbitrage-Free Preprocessing

Before the final option filter is applied, an arbitrage-free representation of the daily market is constructed using [BHK+26]. The default configuration uses surface mode and performs the following operations:

1. Extend the strike grid to improve boundary treatment.
2. Fit a linear (DLV) no-arbitrage option-price curve separately for each expiry.
3. Drop expiries for which a valid fit cannot be obtained.[10]
4. Remove locally redundant strikes associated with negligible fitted density.
5. Fit a surface across strike and expiry to control both butterfly and calendar arbitrage.

At least five strikes per expiry are required. The default minimum fitted density is 10^{-4}, and the minimum admissible ATM volatility is 0.05.

A.5 Option Filtering

The empirical analysis retains observations satisfying

$$\text{BDTE} \leq 70,$$

where BDTE denotes business days to expiry.

The normalized-moneyness filter is

$$-3 \leq z_{\text{ATM}} \leq 2,$$

where z_{ATM} is normalized moneyness computed using the arbitrage-free fitted ATM volatility for the corresponding expiry.

Additional requirements are

$$\begin{aligned} \text{VegaSqrtVar} &\geq 10^{-3}, \\ \log(\text{volume}) &\geq 1, \\ \log(\text{open interest}) &\geq 1. \end{aligned}$$

Here, "VegaSqrtVar" refers to Vega divided by sqrt of expiry. It is therefore the sensitivity to total volatility of the option.

Each retained expiry must contain at least 10 options.

It should be noted that this filter removes many options post 1Y for the early years of the training period due to the minimum volume traded filter. We suspect that this information in IvyDB is not reliable in the past. Future tests might want to remove the volume and/or open interest filters. (It is an option in our data pipeline parametrization.)

A.6 Spot Normalization

Price-like quantities are normalized by the current spot S_d to obtain variables of scale around 1. For example,

$$\widetilde{K}_d^i = \frac{K_d^i}{S_d}, \qquad \widetilde{C}_d^i = \frac{C_d^i}{S_d}, \qquad \widetilde{F}_d^i = \frac{F_d^i}{S_d}.$$

The original spot is retained as $\text{spot_ref}_d = S_d$. (The normalized log-spot level is passed on as h^0 later.)

[10] Existence of such a fit is equivalent to the market not having actionable arbitrage using bid/asks as quoted.

A.7 Arbitrage-free preprocessing and model geometry

The preceding steps produce tradable normalized calls and puts with their observed bid–ask intervals. A per-date arbitrage-free SANOS fit supplies a smooth reference surface, an ATM-volatility curve, and an availability check for the retained quotes. The completed Direct-200 experiments do *not* resample this information onto a secondary reference grid. Instead, every eligible original quote enters the loss at its observed strike and expiry.

The decoder uses the two model-expiry geometries stated in Section **??**: 14 weekly expiries from 5D to 70D, or the same weekly grid augmented by 1D–4D for a total of 18 expiries. On each date it builds one shared support with $N = 200$ inner strikes. Half of the normalized-moneyness candidates cover the first model expiry and half cover the final model expiry over $z \in (-3, 2)$. The candidates are merged and spaced, and the two boundary strikes are

$$\hat{K}_t^0 = \tfrac{1}{2}\hat{K}_t^1, \qquad \hat{K}_t^{N+1} = \tfrac{3}{2}\hat{K}_t^N.$$

The same support is used at every model expiry on that date; only the density and DLV coordinates carry an expiry superscript.

A.8 Fitting ML-SANOS

The native control first fits one unrestricted vector $u_t \in \mathbb{R}^{M(N+1)+2}$ to each date's irregular quote book. The completed affine model then uses the map

$$h_t \in \mathbb{R}^{200} \longmapsto u_t = b + Bh_t \in \mathbb{R}^{M(N+1)+2}$$

from (17). It is initialized from the native controls and then fitted jointly across all 891 dates together with the stable weekly state dynamics. Prices are evaluated directly at every observed quote; the training weights and inside-spread objective are those in (18).

References

[AH11] J. Andreasen and B. Huge. Volatility interpolation. *Risk*, pages 86–89, 2011.

[BGTW19] H. Buehler, L. Gonon, J. Teichmann, and B. Wood. Deep hedging. *Quantitative Finance*, 19(8):1271–1291, 2019.

[BHK+26] Hans Buehler, Blanka Horvath, Anastais Kratsios, Yannick Limmer, and Raeid Saqur. Sanos - smooth arbitrage-free non-parametric option surfaces. https://arxiv.org/abs/2601.11209, Februar 2026.

[BMPW22a] Hans Buehler, Phillip Murray, Mikko S. Pakkanen, and Ben Wood. Deep hedging: learning to remove the drift. *Risk*, February 2022.

[BMPW22b] Hans Buehler, Phillip Murray, Mikko S. Pakkanen, and Ben Wood. Deep hedging: Learning to remove the drift under trading frictions with minimal equivalent near-martingale measures. https://arxiv.org/abs/2111.07844, 2022.

[BR15] Hans Buehler and Evgeny Ryskin. Discrete local volatility for large time steps (extended version). https://papers.ssrn.com/abstract=2642630, 2015.

[BTT07] Aharon Ben-Tal and Marc Teboulle. An old-new concept of convex risk measures: The optimized certainty equivalent. *Mathematical Finance*, 17(3):449–476, 2007.

[Bue06] Hans Buehler. Expensive martingales. *Quantitative Finance*, 6(3):207–218, 2006.

[Bue08] Hans Buehler. Volatility and dividends - volatility modelling with cash dividends and simple credit risk. https://papers.ssrn.com/abstract=1141877, 2008.

[CFD02] Rama Cont, Jose da Fonseca, and Valdo Durrleman. Stochastic models of implied volatility surfaces. *Economic Notes*, 31(2):361–377, 2002.

[CM99] Peter P. Carr and Dilip B. Madan. Option valuation using the fast fourier transform. *Journal of Computational Finance*, 2(4):61–73, 1999.

[CN12] R. Carmona and S. Nadtochiy. Tangent levy market models. *Finance and Stochastics*, 16:63–104, 2012.

[CRW23] Samuel N. Cohen, Christoph Reisinger, and Sheng Wang. Arbitrage-free neural-sde market models. *Applied Mathematical Finance*, 30(1):1–46, 2023.

[Fen09] Matthias R. Fengler. Arbitrage-free smoothing of the implied volatility surface. *Quantitative Finance*, 9(4):417–428, 2009.

[GHL12] Julien Guyon and Pierre Henry-Labordère. Being particular about calibration. *Risk*, January 2012.

[NJZB23] Brian Ning, Sebastian Jaimungal, Xiaorong Zhang, and Maxime Bergeron. Arbitrage-free implied volatility surface generation with variational autoencoders. *SIAM Journal on Financial Mathematics*, 14(4):1004–1027, 2023.

[WBWB19] Magnus Wiese, Lianjun Bai, Ben Wood, and Hans Buehler. Deep hedging: Learning to simulate equity option markets. https://arxiv.org/abs/1911.01700, 2019.

[WWP+21] Magnus Wiese, Ben Wood, Alexandre Pachoud, Ralf Korn, Hans Buehler, Phillip Murray, and Lianjun Bai. Multi-asset spot and option market simulation. https://arxiv.org/abs/2112.06823, 2021.